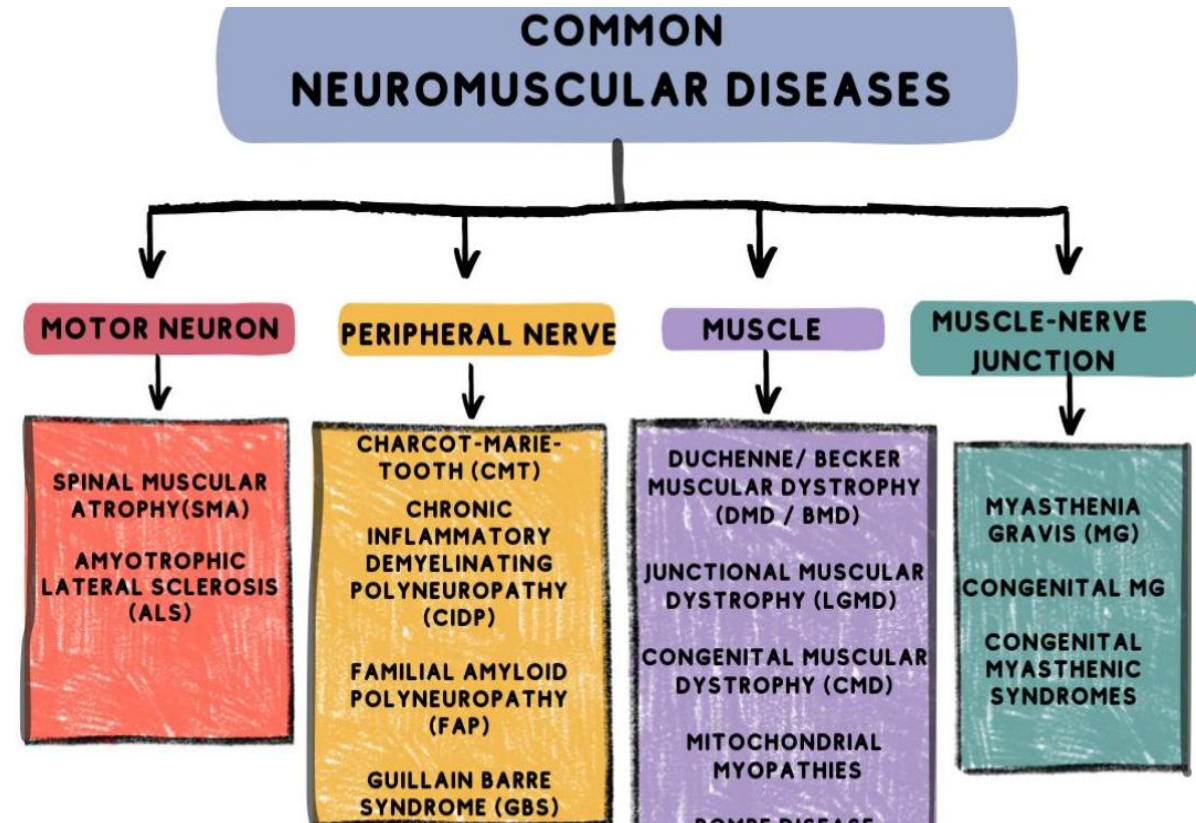


Diseases of Peripheral Nerves and Skeletal Muscles

BY MELESE A. (MD, PATHOLOGIST)

OUTLINES

- Diseases of peripheral nerves
- Diseases of the Neuromuscular Junction
 - Myasthenia Gravis
 - Lambert-Eaton Myasthenic Syndrome
 - Congenital Myasthenic Syndromes
 - Disorders Caused by Toxins
- Diseases of Skeletal Muscle
 - Skeletal Muscle Atrophy

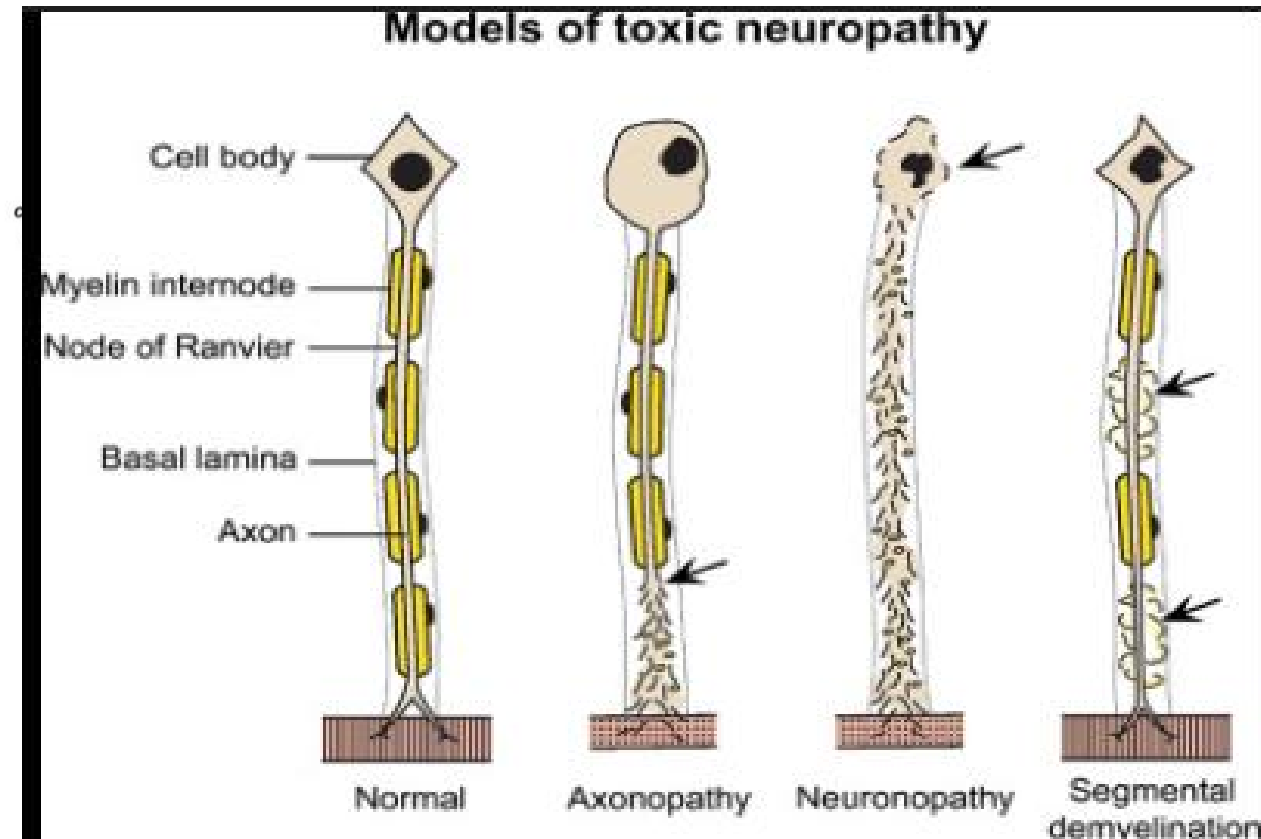


DISEASES OF PERIPHERAL NERVES

General Types of Peripheral Nerve Injury

Axonal Neuropathies

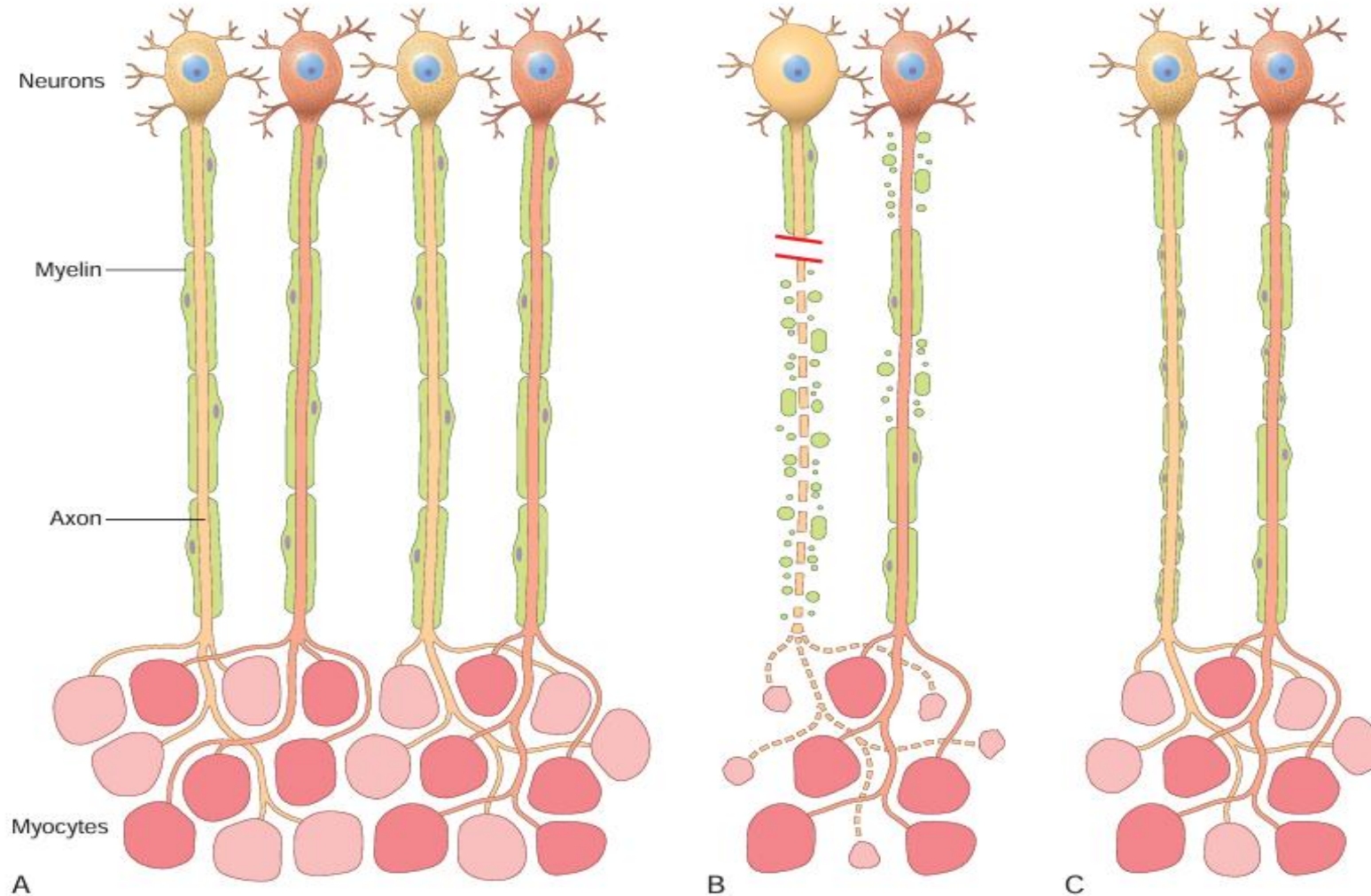
- Axons are the primary target of the damage
- The morphologic hallmarks of axonal neuropathies
 - **Wallerian degeneration**
- Traumatic neuroma
- Causes
 - Trauma
 - toxic and metabolic insults
- The electrophysiologic hallmark
 - **reduction in signal amplitude**
 - relative preservation of conduction velocity



Demyelinating Neuropathies

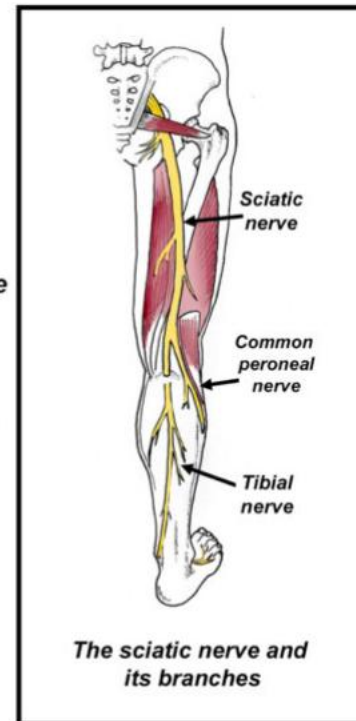
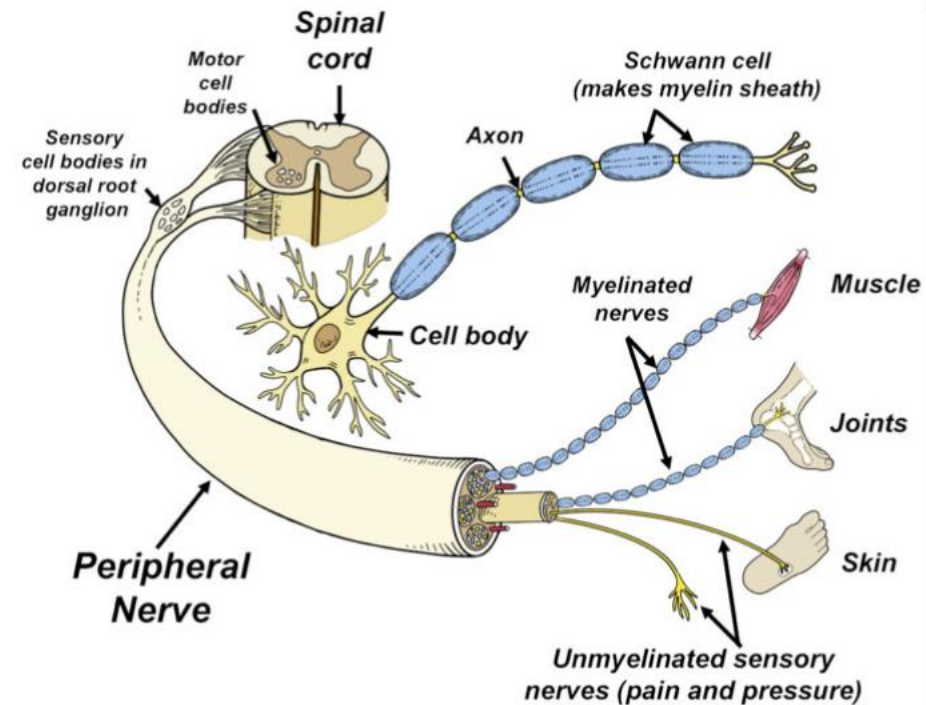
- **Schwann cells** with their **myelin sheaths** are the primary targets of damage
- **Discontinuous** damage of myelin segments
- Repair
 - **shorter and thinner** than the original ones
- The electrophysiologic hallmark
 - **slowed nerve conduction velocity**

Patterns of peripheral nerve damage



Neuronopathies

- Result from destruction of **neurons**
- Herpes zoster and toxins like platinum compounds
- Peripheral nerve dysfunction is equally likely to affect **proximal** and **distal** parts of the body
- **Recovery: Poor**, as there is no cell body to support regeneration.



Anatomic Patterns of Peripheral Neuropathies

Mononeuropathies

- Affect a **single** nerve
- Result in deficits in a **restricted** distribution
- Causes
 - Trauma, entrapment, and infections

Polyneuropathies

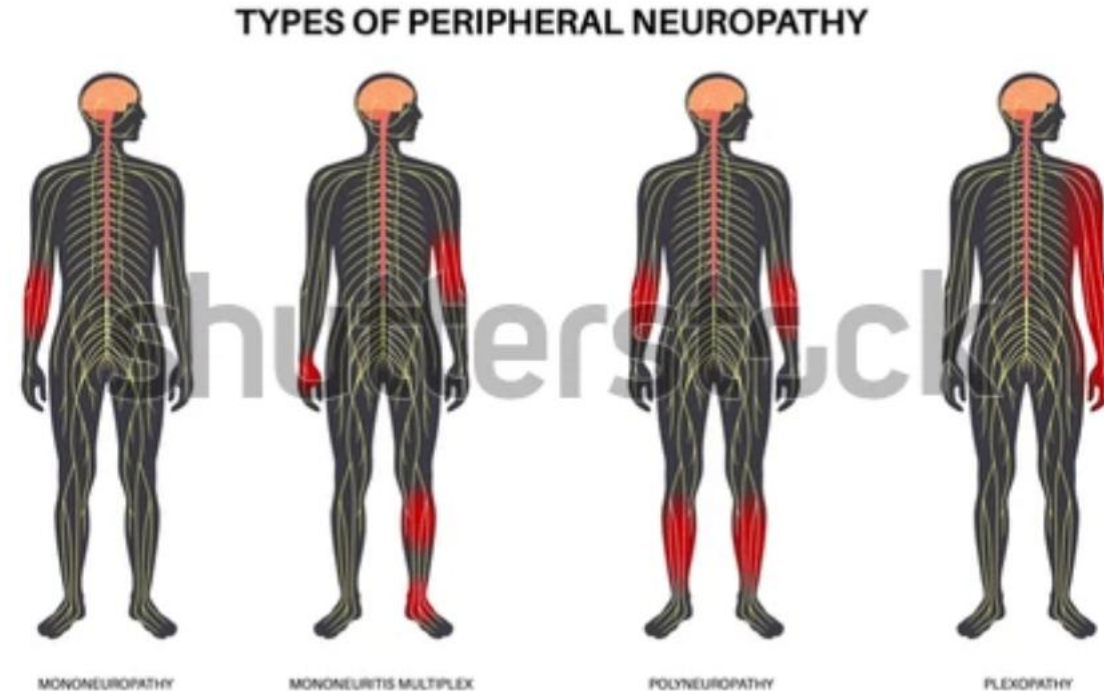
- Involvement of **multiple** nerves
- Deficits start in the feet and ascend with disease progression
- “stocking and glove” distribution

Mononeuritis multiplex

- Damages individual nerves in a **haphazard** fashion
- **Vasculitis** is a common cause

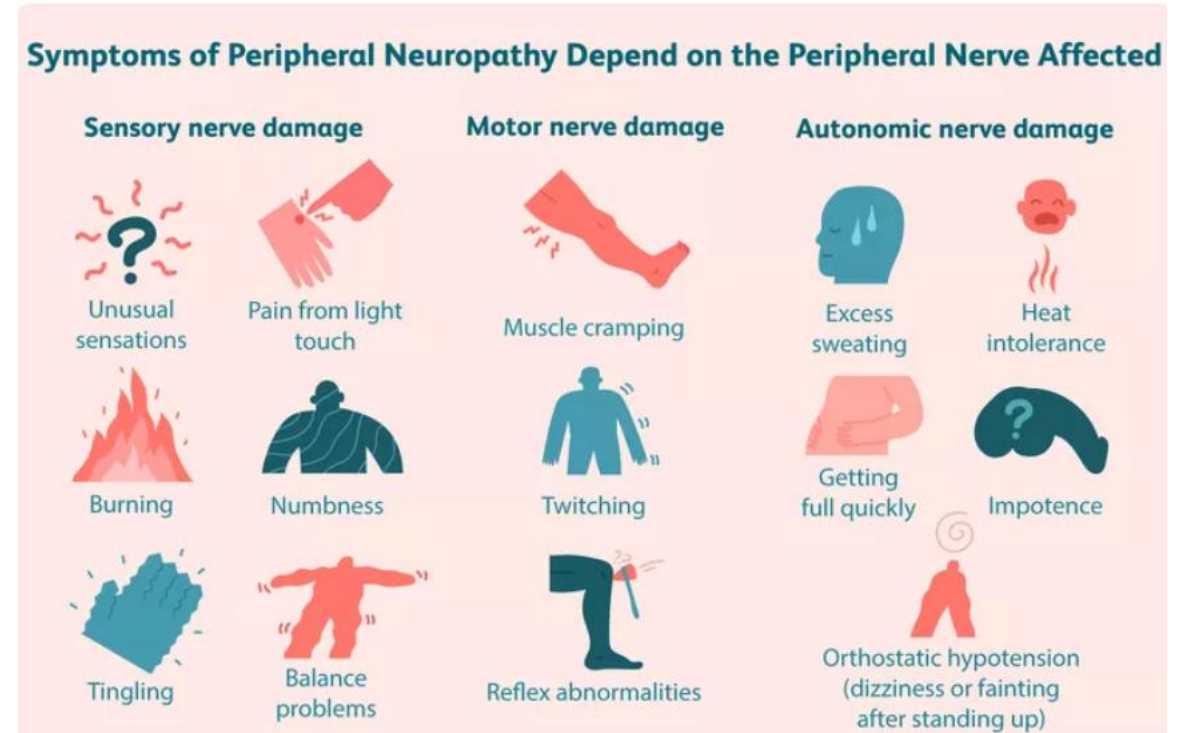
Polyradiculoneuropathies

- Affect nerve roots as well as peripheral nerves
- Lead to **diffuse symmetric symptoms** in **proximal and distal** parts of the body



Peripheral Neuropathies

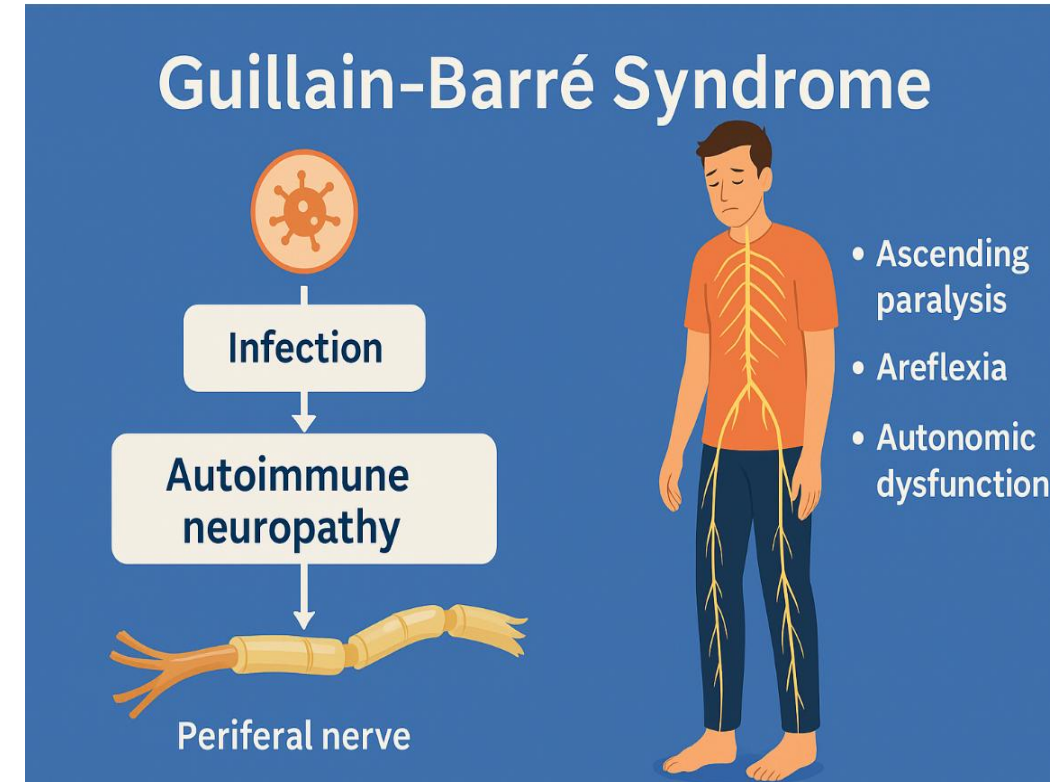
- Patients complain of
 - Numbness
 - painful “pins and needles” sensations
 - weakness
 - in the **distal** portions of the extremities
- Causes
 - Inflammatory diseases
 - Infections
 - Metabolic changes
 - Toxic injury, trauma
 - Paraneoplastic disease
 - Inherited gene defects



Inflammatory Neuropathies

Guillain-Barré Syndrome (Acute Inflammatory Demyelinating Polyneuropathy)

- An immunologically mediated demyelinating peripheral neuropathy
- May lead to life-threatening **respiratory paralysis**
- Characterized clinically by “**ascending paralysis**”
- Histologic features are
 - inflammation
 - demyelination of spinal nerve roots and peripheral nerves (radiculoneuropathy)



Guillain-Barré Syndrome

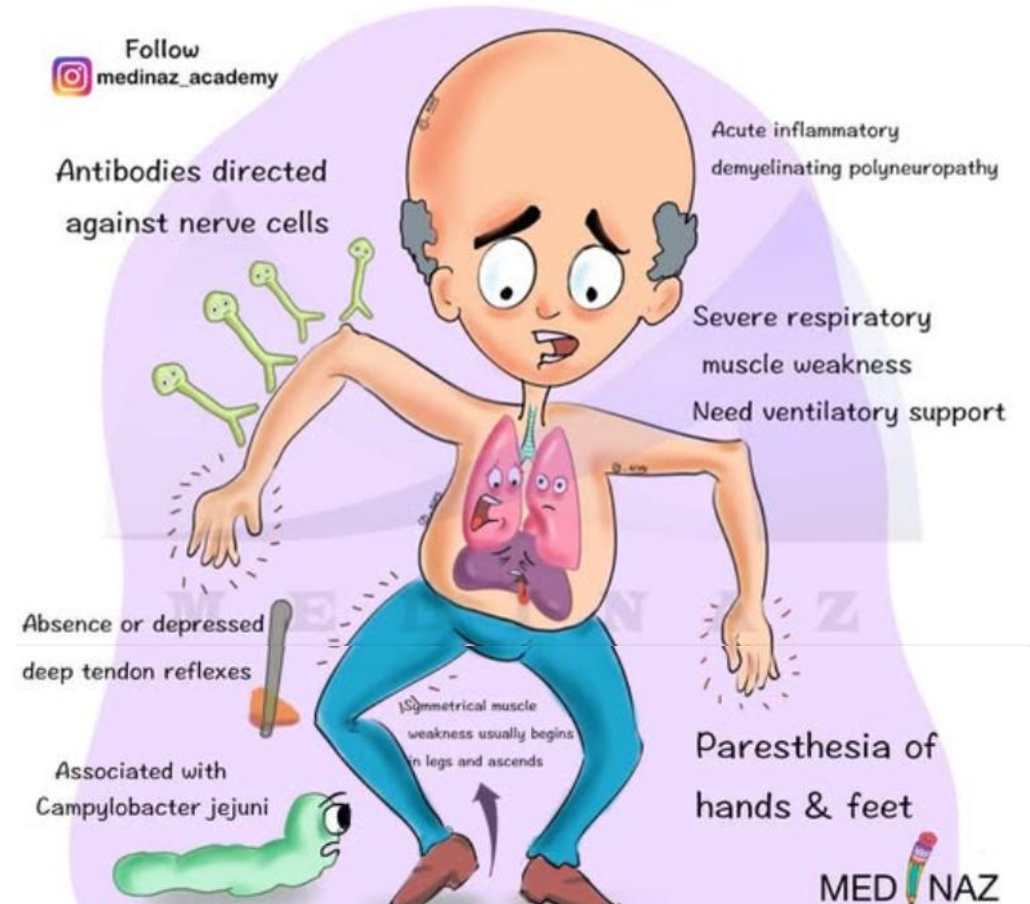
Pathogenesis

- 2/3 of cases are preceded by an acute, influenza-like illness
 - Infections with
 - Campylobacter jejuni, cytomegalovirus
 - Epstein-Barr virus, and Mycoplasma pneumoniae
 - prior vaccination
- Immunologic reaction is favored as the underlying cause

MORPHOLOGY

- Inflammation of peripheral nerves
- Segmental demyelination

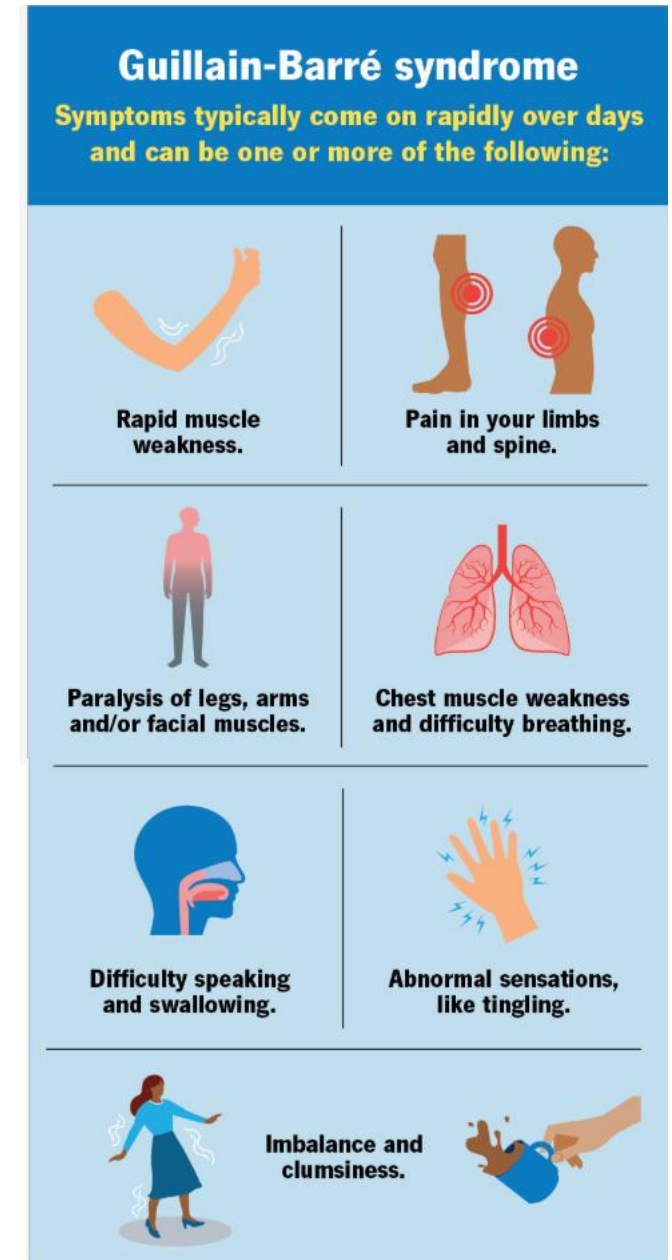
Guillain-Barré syndrome



Guillain-Barré Syndrome

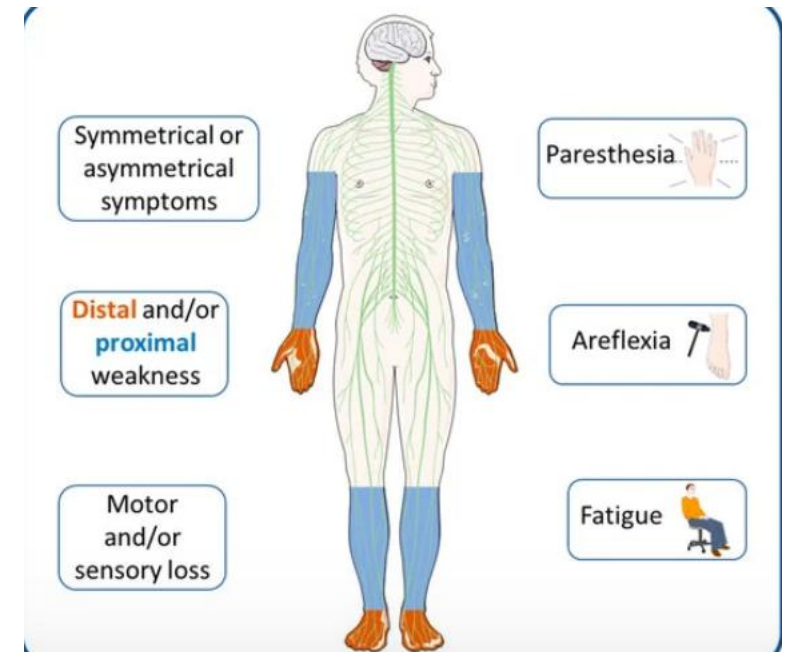
Clinical Features

- The clinical picture is dominated by **ascending paralysis** and **areflexia**.
- Deep tendon reflexes **disappear** early in the process.
- Loss of pain sensation
- Nerve conduction velocities are slowed
- Cerebrospinal fluid (CSF) **protein levels** are **elevated**
- Little or no CSF **pleocytosis**
- Many patients spend weeks in the intensive care unit (ICU)
- **2% to 5%** of affected patients die



Chronic Inflammatory Demyelinating Poly(radiculo)neuropathy

- Characterized by **symmetric** mixed **sensorimotor** polyneuropathy
- Signs and symptoms must be present for at least 2 months
- T cells as well as antibodies are implicated in the inflammatory process.
- **Onion bulbs**



Neuropathy Associated With Systemic Autoimmune Diseases

- Diseases like
 - rheumatoid arthritis
 - Sjögren syndrome
 - systemic lupus erythematosus
- Take the form of **distal sensory or sensorimotor** polyneuropathies.

Neuropathy Associated With Vasculitis

- Noninfectious inflammation of blood vessels
- Vasculitis often presents as **mononeuritis multiplex**
- Peripheral neuropathy that occurs in systemic vasculitis(MPO-ANCA–associated vasculitides)
 - microscopic polyangiitis
 - eosinophilic granulomatosis with polyangiitis/Churg-Strauss syndrome)
 - polyarteritis nodosa
 - granulomatosis with polyangiitis(PR3-ANCA–associated)
- Peripheral nerves typically show **patchy axonal degeneration and loss**

Infectious Neuropathies

Leprosy (Hansen Disease)

- Peripheral nerves are involved in both **lepromatous** and **tuberculoid** leprosy
- **Lepromatous leprosy**
 - Segmental demyelination and remyelination
 - Affected individuals develop a **symmetric polyneuropathy**
 - most severe in the relatively cool distal extremities and in the face
 - The infection prominently involves **pain fibers**
- **Tuberculoid leprosy**
 - usually manifests as dermal nodules
 - affected individuals have much more localized nerve involvement.



Infectious Neuropathies

HIV/AIDS

- Early-stage HIV infection can be associated with mononeuritis multiplex and demyelinating disorders
- Later stages of HIV infection are associated with a **distal sensory neuropathy**

Diphtheria

- From the effects of the diphtheria exotoxin
- Produces an **acute peripheral neuropathy** associated with prominent **bulbar and respiratory muscle dysfunction**

Varicella-Zoster Virus

- One of the most common **viral** infections of the peripheral nervous system
- Following chickenpox
 - **Latent** infection persists within **neurons of sensory ganglia**
 - Painful, vesicular skin eruption in a distribution that follows sensory dermatomes (**shingles**).
 - **thoracic or trigeminal nerve dermatomes**

Metabolic, Hormonal, and Nutritional Neuropathies

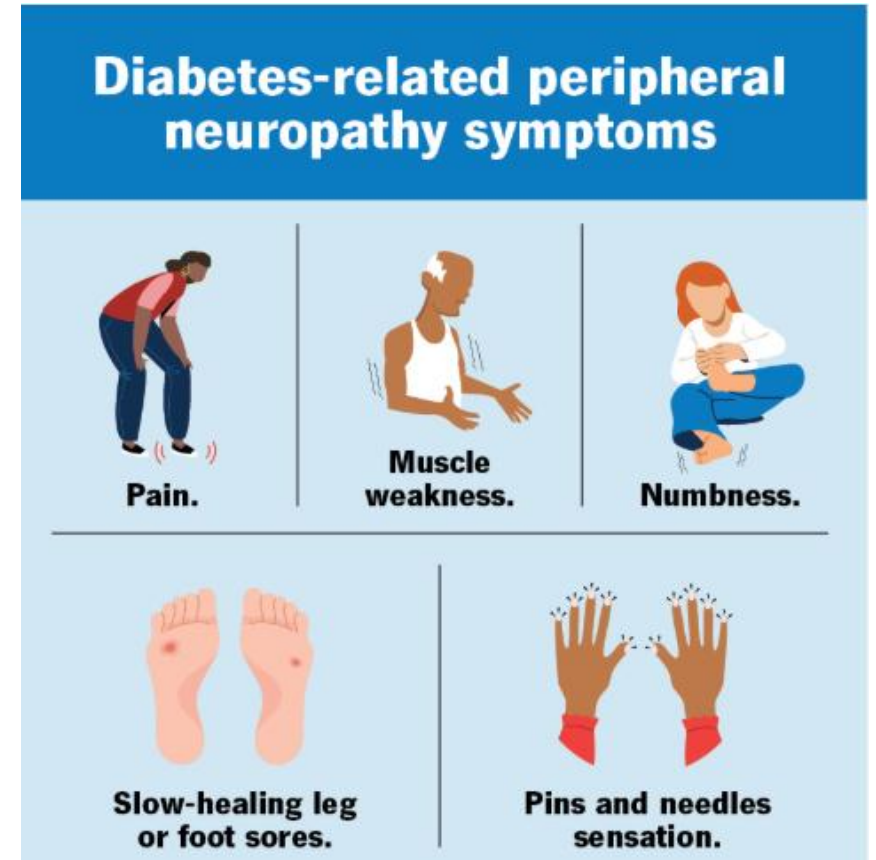
Diabetes

- The most common cause of peripheral neuropathy
- Prevalence depends on the duration of the disease
- Patients with both type 1 and type 2 diabetes are affected
- Several distinct clinicopathologic patterns
 - an ascending distal symmetric sensorimotor polyneuropathy
 - axonal neuropathy
- **Pathogenesis**
 - both metabolic and secondary vascular changes
 - Hyperglycemia causes the nonenzymatic **glycosylation** of proteins, lipids, and nucleic acids
 - advanced glycosylation end-products (AGEs) : interfere with normal protein function
 -

Diabetes

Clinical Features

- Distal symmetric diabetic polyneuropathy
 - numbness, loss of pain sensation,
 - difficulty with balance,
 - paresthesias or dysesthesias
- Neuropathy leads to
 - increased susceptibility to foot and ankle fractures
 - chronic skin ulcers
- Dysfunction of the autonomic nervous system
 - postural hypotension, incomplete emptying of the bladder, sexual dysfunction
- mononeuropathy, cranial neuropathy, and **radiculoplexus neuropathy**



Other Metabolic, Hormonal, and Nutritional Neuropathies

Uremic neuropathy

- individuals with renal failure
- distal, symmetric neuropathy

Thyroid dysfunction

- Hypothyroidism
 - carpal tunnel syndrome
- **Hyperthyroidism**
 - neuropathy resembling Guillain-Barré syndrome

Vitamin B12 (cyanocobalamin) deficiency

- subacute combined degeneration with damage to long tracts in the spinal cord

Deficiencies of vitamin B1 (thiamine), vitamin B6 (pyridoxine), folate, copper, and zinc

Toxic Neuropathies

- Important causes of **toxic** peripheral nerve damage include
 - alcohol (independent of associated nutritional deficiencies),
 - heavy metals (lead, mercury, arsenic, and thallium), and
 - organic solvents
 - Various medications
 - chemotherapeutic agents

Neuropathies Associated With Malignancy

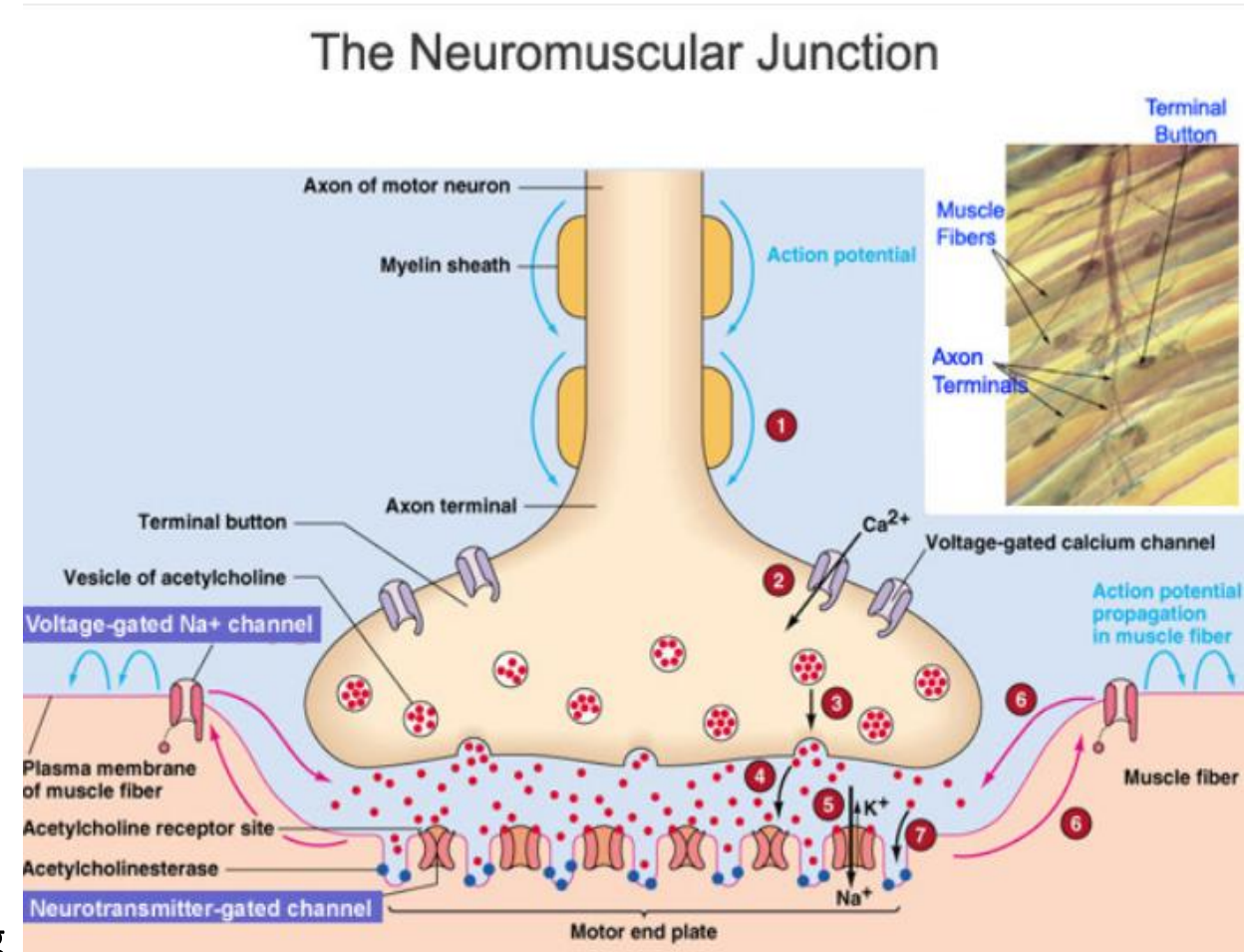
- Direct infiltration or compression of peripheral nerves
- Paraneoplastic neuropathies
- Neuropathies associated with monoclonal gammopathies

Neuropathies Caused by Physical Forces

- Lacerations
- Avulsion
- Compression neuropathy (entrapment neuropathy)
 - Carpal tunnel syndrome
 - ulnar nerve at the level of the elbow
 - peroneal nerve at the level of the knee
 - radial nerve in the upper arm (“Saturday night palsy”)

Diseases of the Neuromuscular Junction

- The neuromuscular junction is a complex specialized structure
 - interface of motor nerve axons and skeletal muscle
 - serves to control muscle contraction
- They are found midway along the length of myofibers.
 - bulbous synaptic boutons.
- Upon depolarization, these presynaptic nerve terminals release acetylcholine (ACh) into the **synaptic cleft**
- The postsynaptic sarcolemma
 - acetylcholine receptors (AChR).
 - responsible for the initiation of signals leading to muscle contraction.

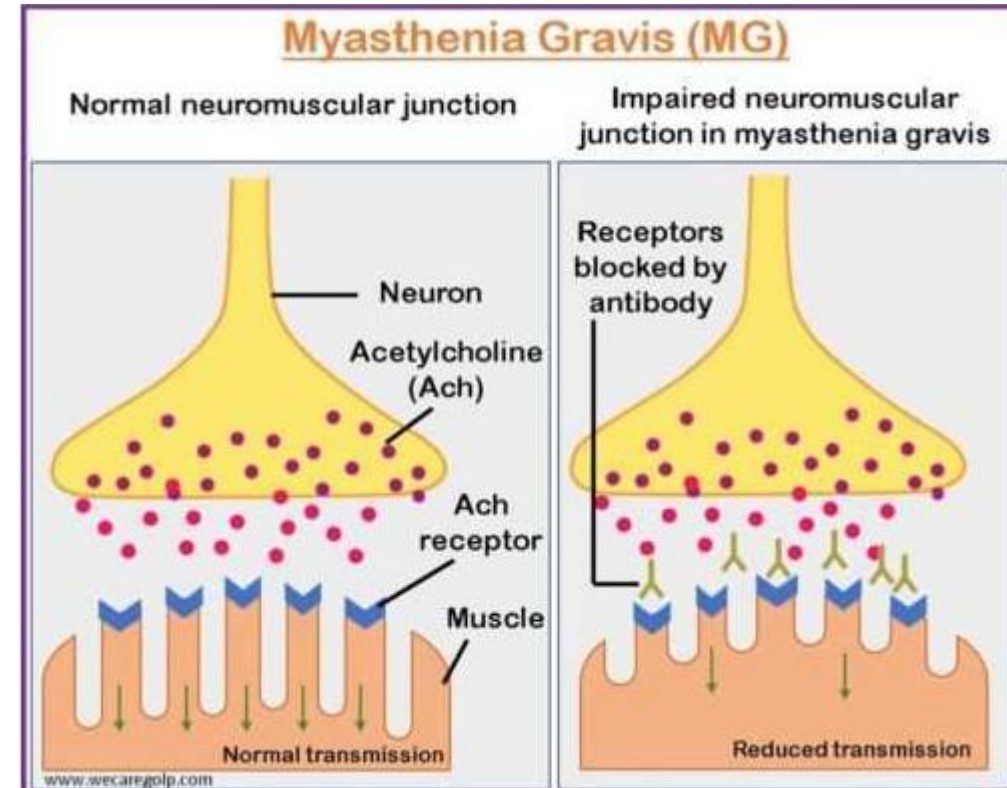


- Regardless of cause, disorders that impair the function of neuromuscular junctions tend to present with **painless weakness**.
 - **Autoantibodies** that inhibit key neuromuscular junction proteins
 - The most common cause of disrupted neuromuscular transmission
 - myasthenia gravis (grave weakness).
 - **Inherited defects** in specialized neuromuscular junction proteins
 - associated with **myasthenic syndromes**
 - Disorders caused by **toxins** that alter neuromuscular transmission
 - rarely encountered

Antibody-Mediated Diseases

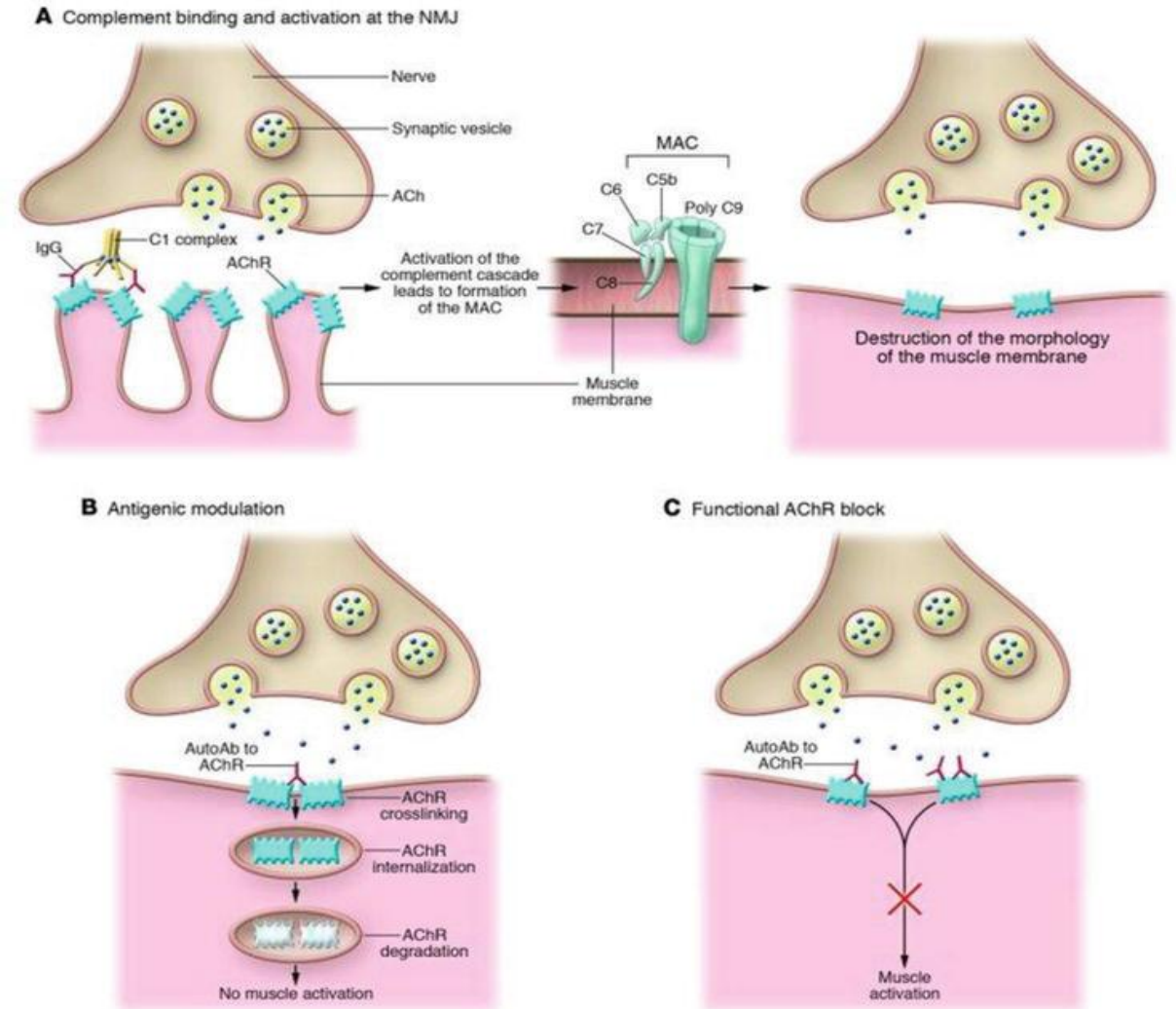
Myasthenia Gravis

- Is an **autoimmune** disease
- Is usually associated with **autoantibodies** directed against **acetylcholine receptors**.
- It has a prevalence of 150 to 200 per million
- Shows a **bimodal** age distribution.
 - The female-to-male ratio is 2 : 1 in the **young adults**, but in older adults there is a male predominance



Pathogenesis

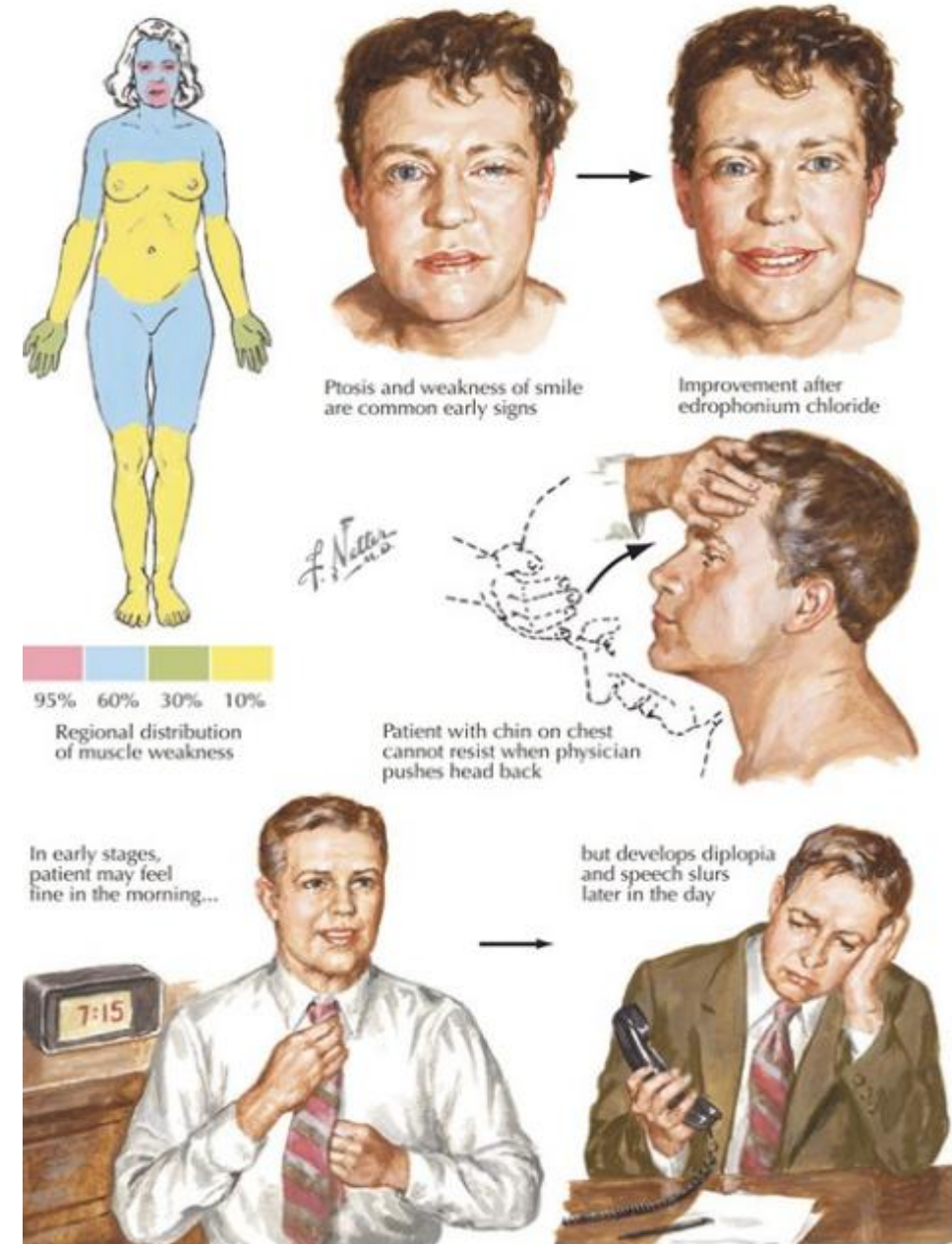
- **Anti-acetylcholine receptor antibodies(85%)**
 - lead to the **aggregation** and **degradation** of the receptors
 - damage the postsynaptic membrane through **complement fixation**.
 - depleted acetylcholine receptors
 - This limits the ability of myofibers to respond to acetylcholine
- **Autoantibodies directed against muscle-specific receptor tyrosine kinase**
 - interfere with the trafficking and clustering of acetylcholine receptor
 - the net effect is **decreased acetylcholine receptor function**.



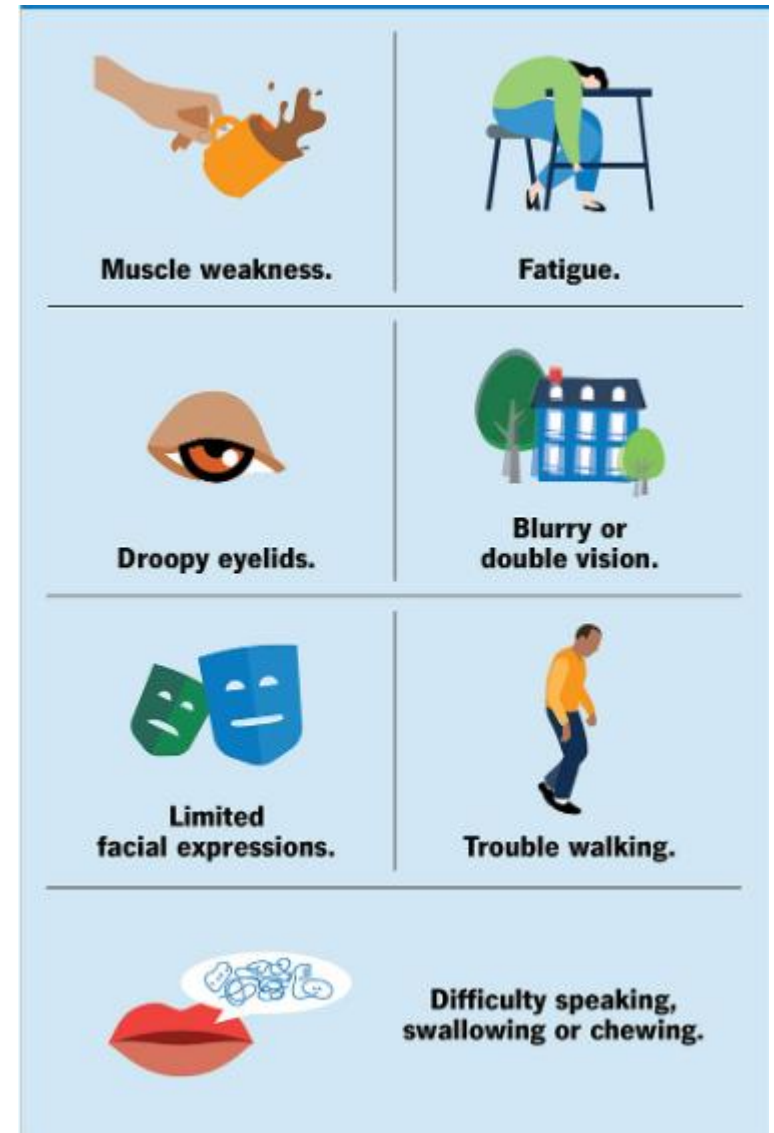
- There is a strong association with **thymic abnormalities**.
 - 10% of patients have a **thymoma**
 - tumor of thymic epithelial cells
 - 30% of patients (particularly young) have **thymic hyperplasia**
 - B-cell follicles in the thymus

Clinical Features

- Patients with anti-acetylcholine receptor antibodies
 - Present with **fluctuating weakness**
 - worsens **with exertion** and often **over the course of the day**.
 - **Diplopia** and **ptosis** due to involvement of **extraocular muscles** are common
 - distinguish myasthenia gravis from myopathies
 - In some patients
 - symptoms are **confined to ocular muscles**
 - others develop **generalized weakness**
- Cases with antibodies against muscle-specific receptor tyrosine kinase
 - exhibit more focal muscle involvement
 - neck, shoulder, facial, respiratory, and bulbar muscles

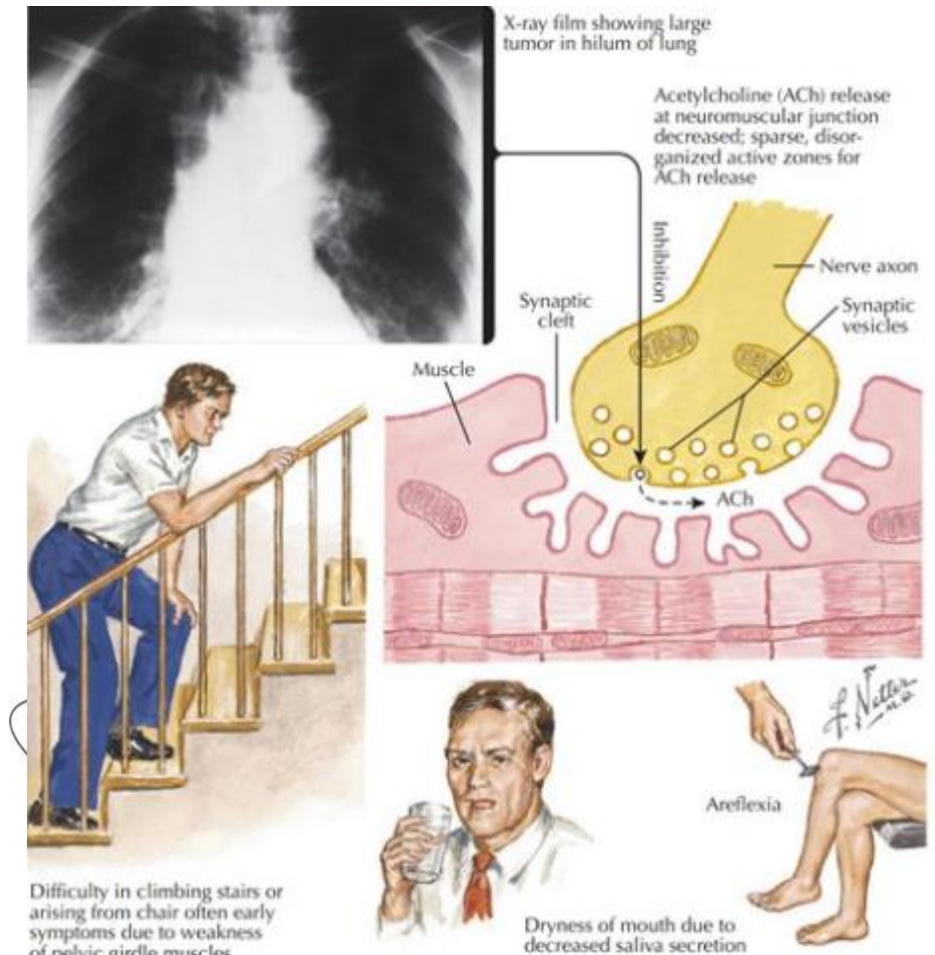


- Diagnosis is based on
 - clinical history
 - physical findings
 - the identification of autoantibodies
 - electrophysiologic studies.
- Treatment
 - Acetylcholinesterase inhibitors that increase the half-life of acetylcholine
 - plasmapheresis and immunosuppressive drugs
 - Thymectomy



Lambert-Eaton Myasthenic Syndrome

- Is an autoimmune disorder caused by antibodies
 - Block **acetylcholine release** by inhibiting a **presynaptic calcium channel**.
- **Rapid repetitive stimulation** increases muscle response.
 - Muscle strength is augmented after a few seconds of muscle activity.
- Patients typically present with **weakness of their extremities**.
- In about **half** of cases there is an **underlying malignancy**,
 - most often **neuroendocrine carcinoma** of the lung.
 - Symptoms may precede the diagnosis of cancer
- The stimulus for autoantibody formation in paraneoplastic case
 - expression of the **same calcium channel** in the neoplastic cells.
- Patients **without cancer** often have other autoimmune diseases:
 - vitiligo or thyroid disease.
- Treatment consists of
 - drugs that increase acetylcholine release by depolarizing synaptic membranes
 - immunosuppressive agents



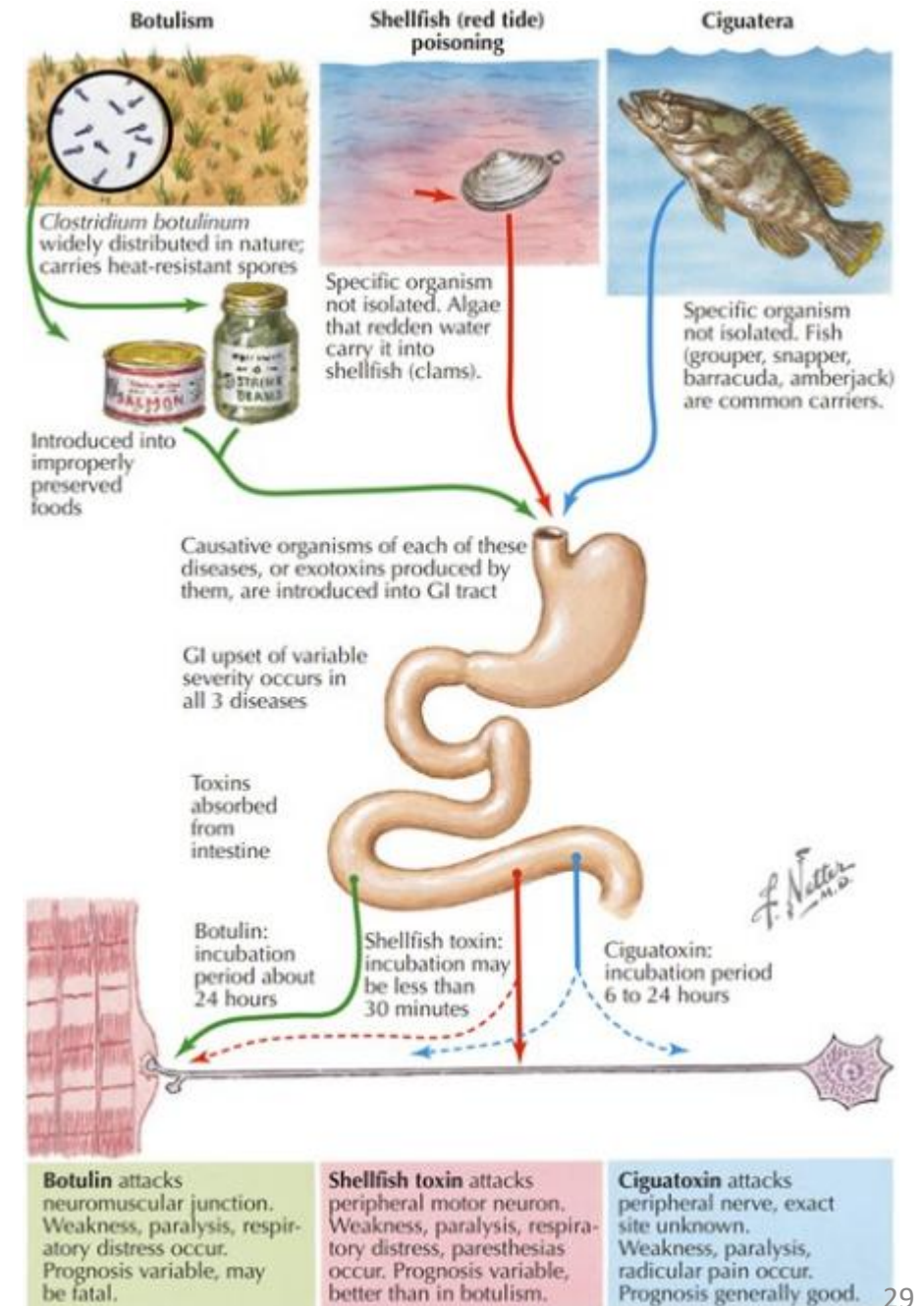
Congenital Myasthenic Syndromes

- Have an **autosomal recessive** mode of inheritance
- Marked by varying degrees of muscle weakness.
- Causative mutations in genes encoding
 - presynaptic, synaptic, or postsynaptic proteins.
- Many patients with congenital myasthenic syndromes present
 - in the **perinatal period**
 - with poor muscle tone, external eye muscle weakness, and breathing difficulties
 - others have milder forms of the disease and may not come to clinical attention until adolescence or adulthood.
- The clinical presentation, response to drugs such as acetylcholinesterase inhibitors, and prognosis depend largely on the underlying mutation.

Disorders Caused by Toxins

Botulism

- A **toxin-mediated** disease causing **acute, rapid-onset** presynaptic failure.
- **Pathophysiology:** *Clostridium botulinum* neurotoxin (BoNT) enters nerve terminals and **cleaves proteins essential for ACh vesicle fusion**, causing **profoundly reduced release**.
- **Key Features:**
 - **Acute, descending flaccid paralysis** starting with cranial nerves (blurred vision, dysphagia, dysarthria, ptosis).
 - Progresses to symmetric limb and respiratory muscle weakness.
 - **Autonomic symptoms** (dry mouth, ileus, urinary retention, pupillary dilation).
 - **No sensory deficits.**



Diseases of Skeletal Muscle

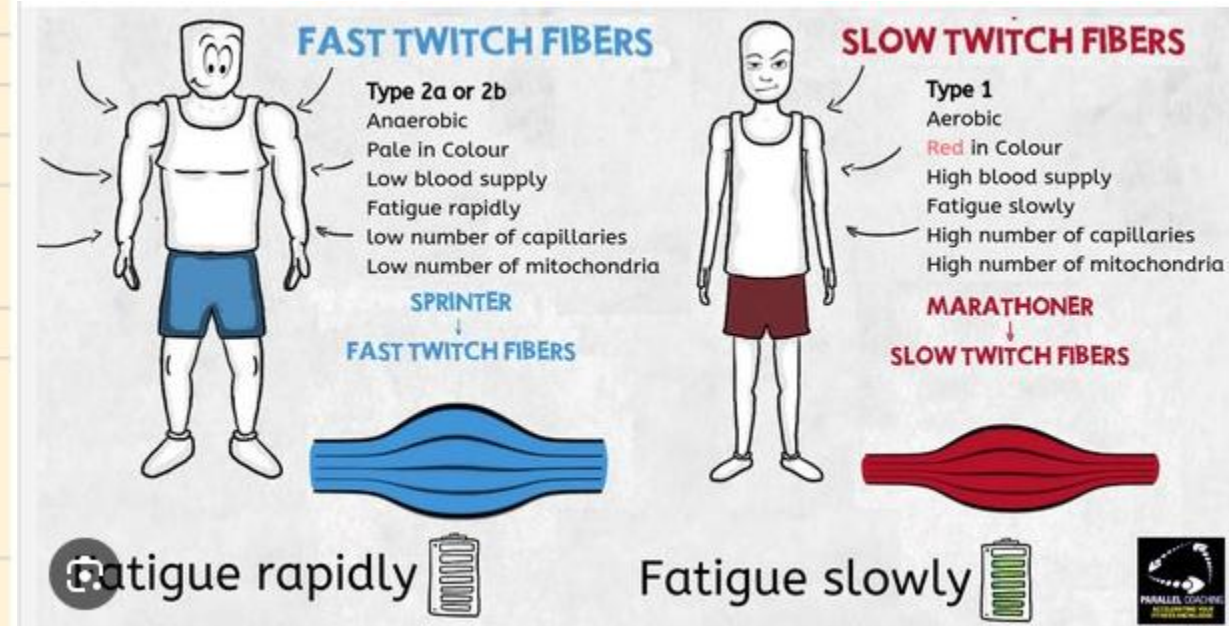
INTRODUCTION

- Skeletal muscle has unique structural, cellular, and molecular characteristics and unique patterns of injury and repair.
- Myofibers are arranged in fascicles, each associated with a small pool of tissue **stem cells** referred to as **satellite cells**,
 - which can contribute to **muscle regeneration** following injury.
- Myofibers are of two main types, **type I** and **type II**, which are admixed in a **checkerboard pattern** in normal skeletal muscle.
- Fiber type is determined by signals received from innervating motor neurons and, as a result, all fibers that are part of a motor unit are of the **same type**.

Table 27-1 Muscle fiber types

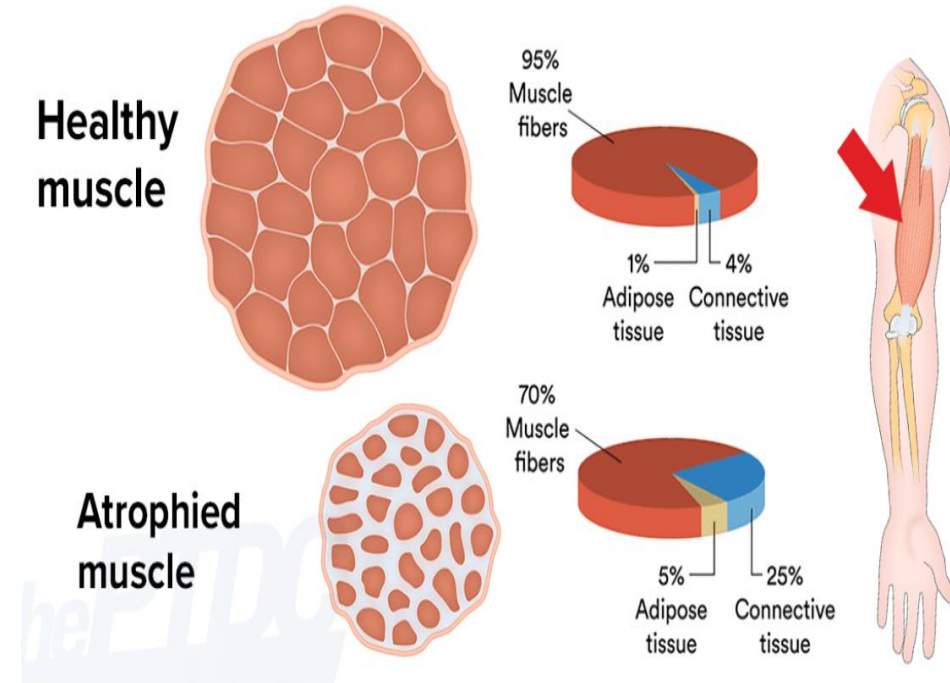
	Type I	Type II
Action	Sustained force	Fast movement
Activity type	Aerobic exercise	Anaerobic exercise
Power produced	Low	High
Resistance to fatigue	High	Low
Lipid content	High	Low
Glycogen content	Low	High
Energy metabolism	Low glycolytic capacity, high oxidative capacity	High glycolytic capacity, low oxidative capacity
Mitochondrial density	High	Low
Enzyme activity	NADH-TR, dark staining ATPase at pH 4.3, dark staining ATPase at pH 9.4, light staining	NADH-TR, light staining ATPase at pH 4.3, light staining ATPase at pH 9.4, dark staining
Myosin heavy chain gene expressed	<i>MYH7</i>	<i>MYH2, MYH4, MYH1</i>
Color	Red (high myoglobin content)	Pale red / tan (low myoglobin content)
Prototype	Soleus (pigeon)	Pectoral (pigeon)

ATPase, Adenosine triphosphatase; NADH-TR, nicotinamide adenine dinucleotide, reduced form, tetrazolium reductase.



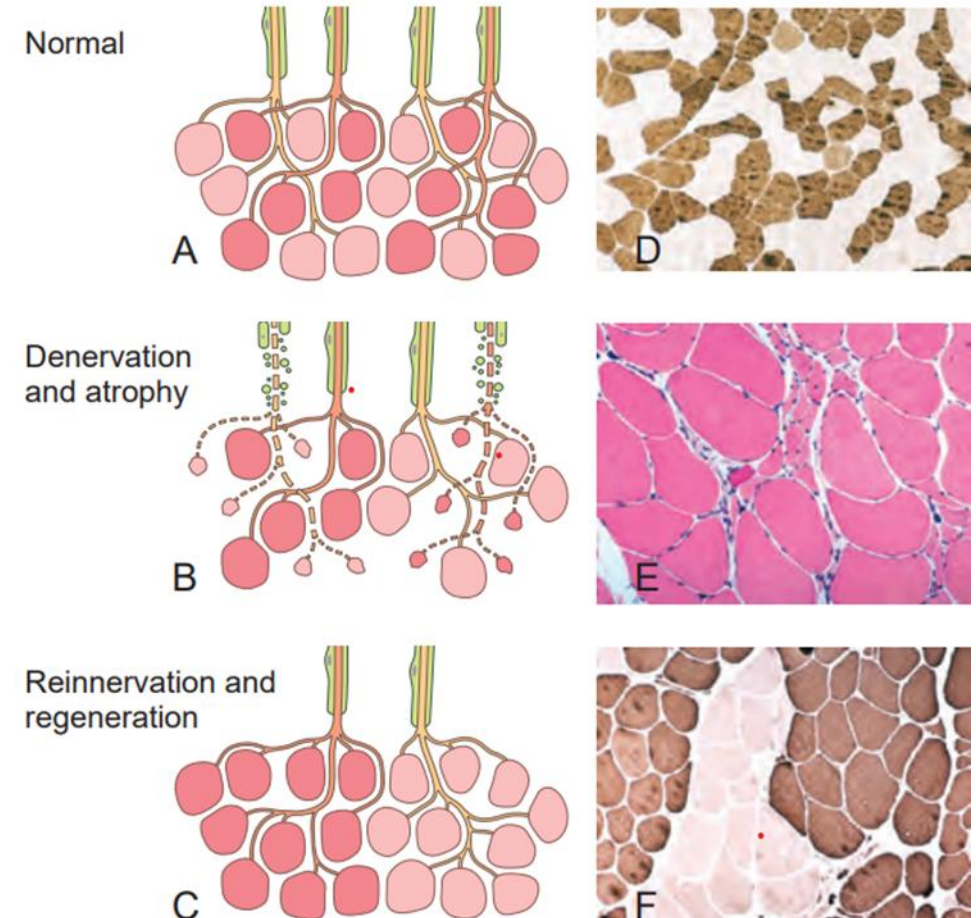
Skeletal Muscle Atrophy

- Skeletal muscle atrophy is a common feature of many disorders.
 - Loss of innervation, disuse, cachexia, old age, and primary myopathies
- Certain patterns of atrophy are suggestive of specific underlying etiologies:
 - Clusters or groups of atrophic fibers are seen in **neurogenic disease**
 - Perifascicular atrophy is seen in **dermatomyositis**
 - Type II fiber atrophy with sparing of type I fibers is seen with **prolonged corticosteroid therapy or disuse**



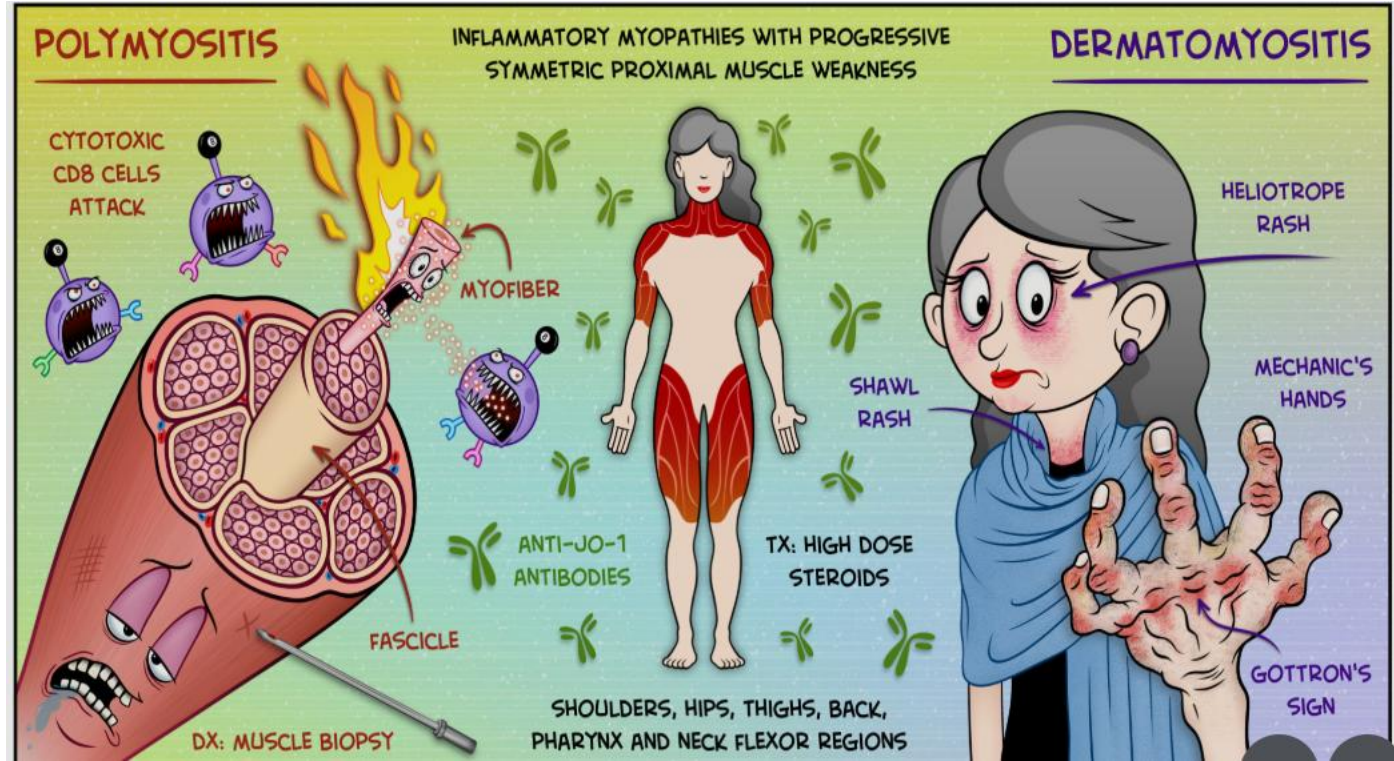
Neurogenic and Myopathic Changes in Skeletal Muscle

- Disorders impacting skeletal muscle
 - **myopathic injury** (damaging myofibers directly)
 - **neurogenic injury** (disrupting muscle innervation)
 - May lead to fiber type grouping and grouped atrophy
- Muscle fiber type is determined by the **innervating motor neuron**
- Morphologic changes associated with primary myopathic processes
 - Segmental myofiber degeneration and regeneration
 - When only part of a myofiber undergoes necrosis
 - Degeneration=> Raised levels of creatine kinase
 - Myofiber hypertrophy
 - physiologic adaptation to exercise or in association with certain chronic myopathic conditions.
 - Cytoplasmic inclusions in the form of vacuoles, aggregates of proteins, or clustered organelles
 - characteristic of several primary forms of myopathy.



Inflammatory Myopathies

- Polymyositis
- Dermatomyositis
- Inclusion body myositis
- Immune-mediated disorders
 - systemic lupus erythematosus
 - systemic sclerosis
 - sarcoidosis
 - infectious agents



Dermatomyositis

- Is a systemic autoimmune disease that typically presents with
 - proximal muscle weakness
 - skin changes

Pathogenesis

- Damage to **small blood vessels** contributes to muscle injury.
- Biopsies of muscle and skin may show deposition of the complement membrane attack complex (**C5b-9**) within capillary beds in both tissues.
- B lymphocytes as well as plasma cells are part of the inflammatory infiltrate that is seen in muscles.
- Certain autoantibodies tend to be associated with specific clinical features:
 - Anti-Mi2 antibodies
 - Anti-Jo1 antibodies
 - Anti-P155/P140 antibodies

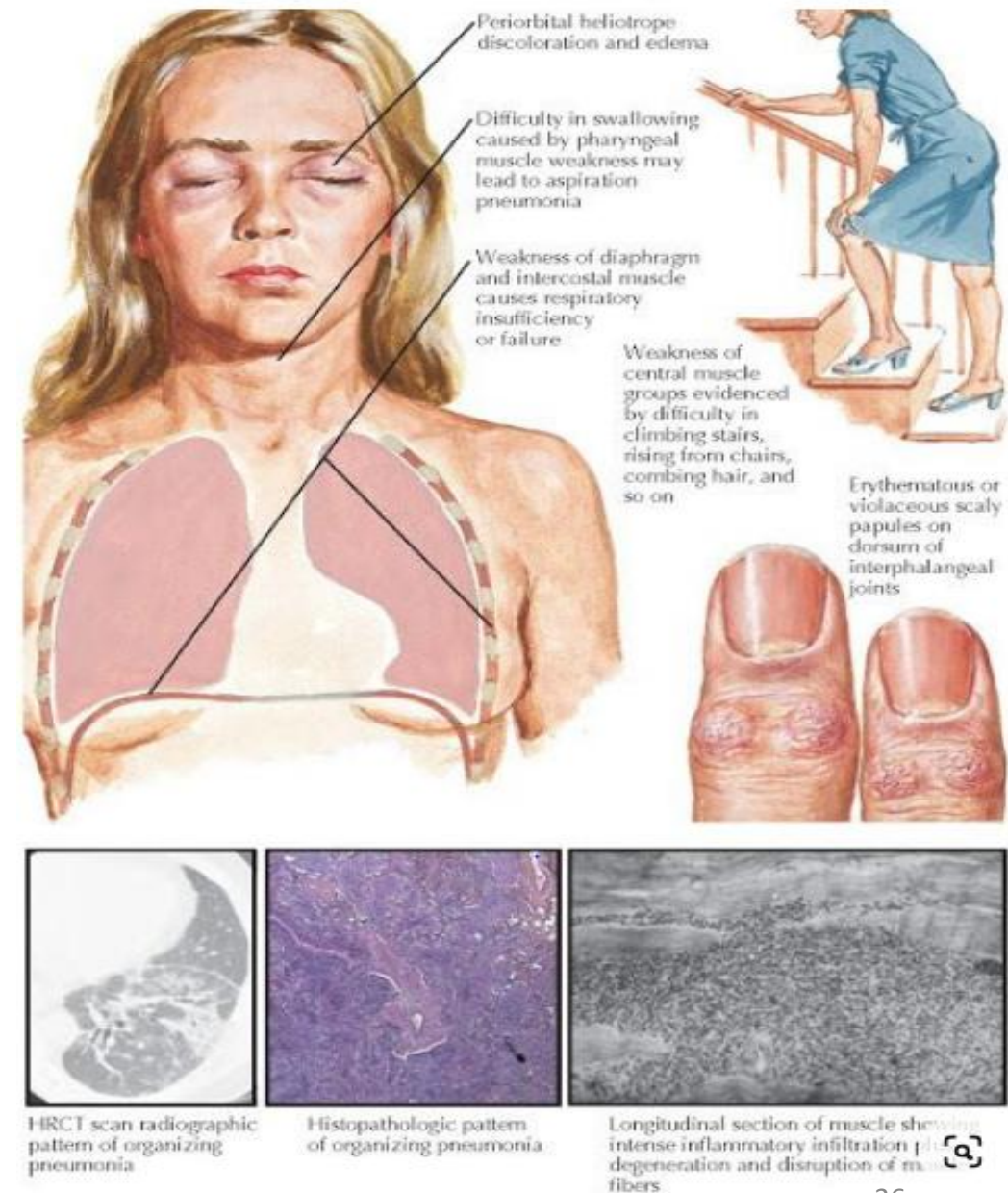
Dermatomyositis

MORPHOLOGY

- Infiltrates of mononuclear inflammatory cells in
 - **Perimysial** connective tissue and around blood vessels
- Myofiber atrophy accentuated at the edges of the **fascicles**—**perifascicular atrophy**
- Segmental fiber necrosis and regeneration may also be seen
- IHC-CD4+ T-helper cells and the deposition of C5b-9 in capillary vessels

Clinical Features

- Muscle weakness is slow in onset, symmetric, and often accompanied by myalgias.
 - It typically affects the **proximal muscles** first.
- Elevated serum **creatine kinase** levels
- Lilac colored discoloration of the upper eyelids (**heliotrope rash**)
- Scaling erythematous eruption or dusky red patches over the knuckles, elbows, and knees (**Gotttron papules**)
- Dysphagia (one third of patients)
- Interstitial lung disease (10%)
- Cardiac involvement



- Juvenile and adult forms are recognized.
- The average age of onset
 - juvenile dermatomyositis is 7 years
 - adult cases present from the 4th to 6th decade of life.
- Dermatomyositis is the most common inflammatory myopathy in children.
- Childhood disease is more likely to be associated with calcinosis and lipodystrophy
- The overall prognosis is better in children than in adults.
- 15% to 24% of adult patients have an associated **malignancy**.

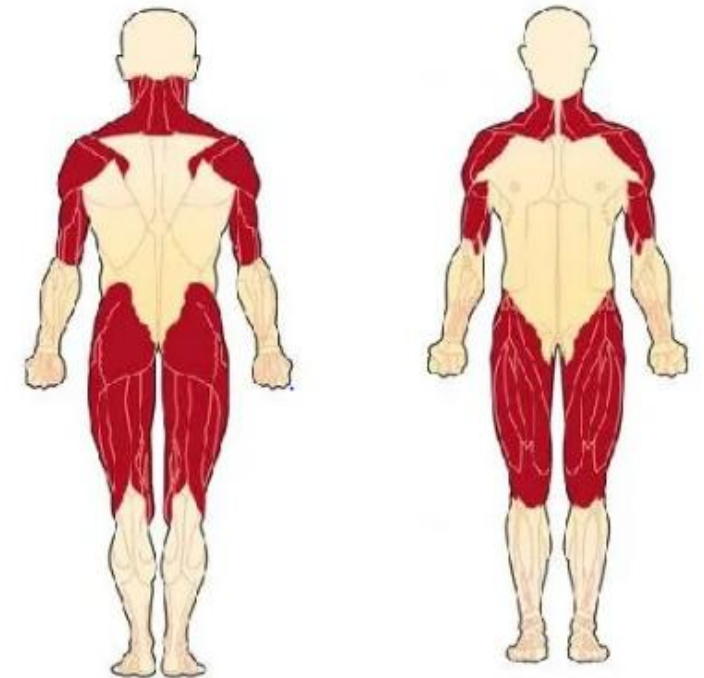
Polymyositis

- An adult -onset inflammatory myopathy that shows myalgia and weakness
 - lacks cutaneous features
 - a diagnosis of exclusion.
- Patients typically develop
 - **symmetric proximal** muscle involvement
 - there may be inflammatory involvement of the heart and the lungs, as well as similar autoantibodies.

Pathogenesis

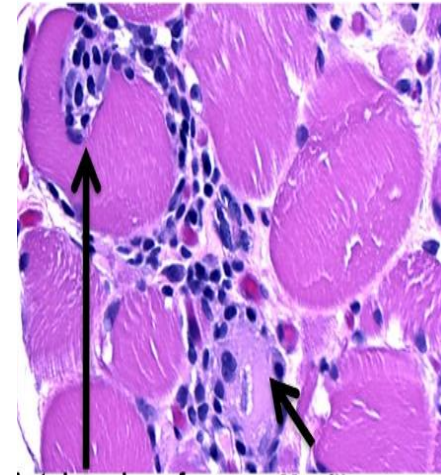
- The pathogenesis of polymyositis is uncertain, but it is believed to have an immunologic basis.
 - CD8-positive cytotoxic T cells
 - vascular injury is not believed to have a major role in polymyositis.
- Mononuclear inflammatory cell infiltrates(endomysial)

polymyositis



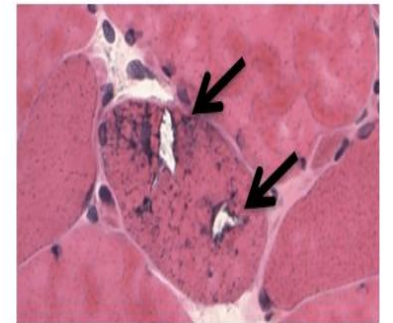
Inclusion Body Myositis

- Disease of late adulthood
- The most common inflammatory myopathy in patients older than age 65 years.
- Most affected individuals present with slowly progressive muscle weakness
 - tends to be most severe in the quadriceps and the distal upper extremity muscles.
- Dysphagia from esophageal and pharyngeal muscle involvement is **not uncommon**.
- Laboratory studies usually show modestly elevated creatine kinase levels
- Most myositis-associated autoantibodies are absent



Autoinvasion of
Mononuclear Cells

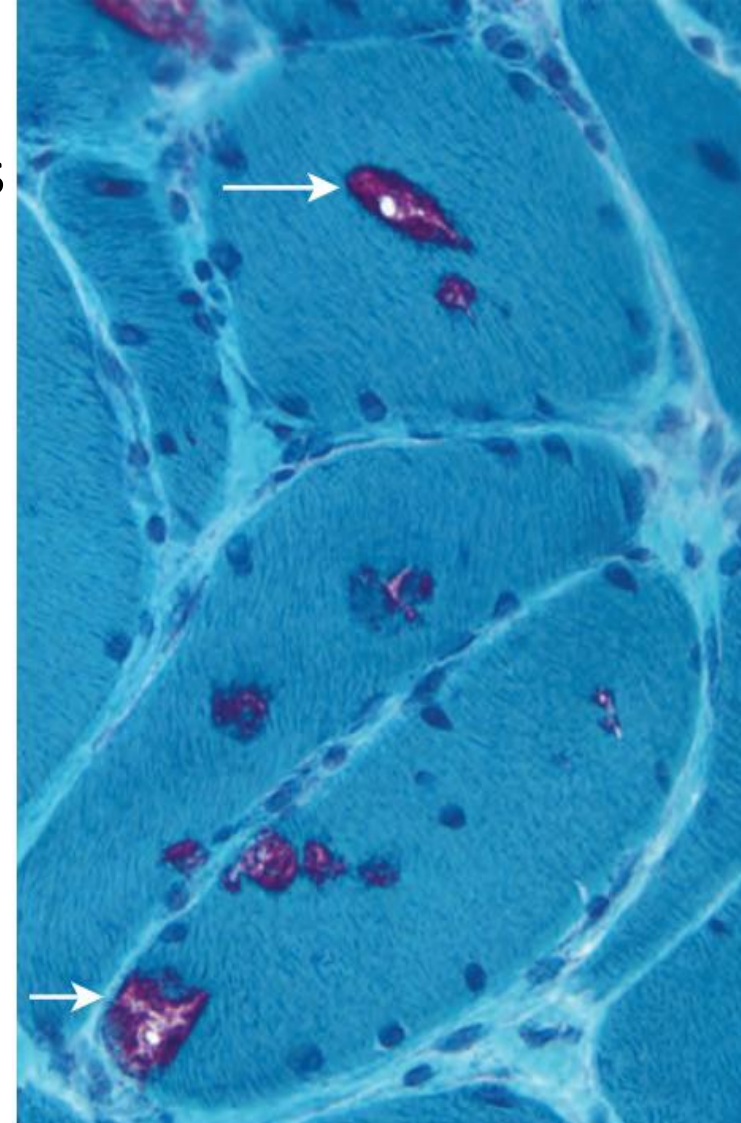
Myofiber
Degeneration



Rimmed vacuoles (RVs) and protein aggregates

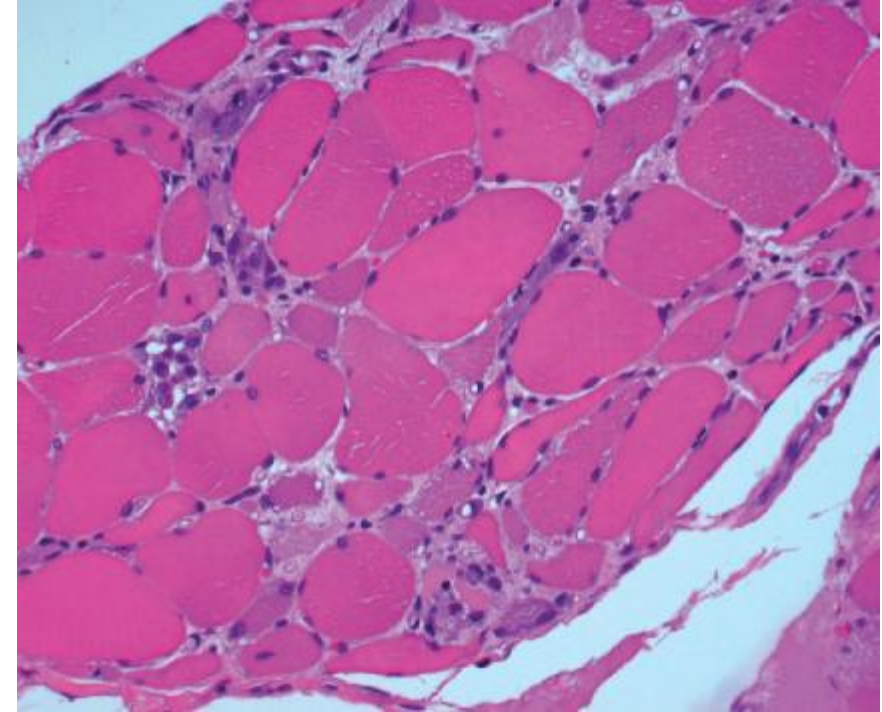
MORPHOLOGY

- Abnormal cytoplasmic inclusions described as **“rimmed vacuoles”**
- Tubulofilamentous inclusions in myofibers,
 - seen by electron microscopy
- Cytoplasmic inclusions containing proteins typically associated with neurodegenerative diseases, like
 - beta-amyloid, TDP-43, and ubiquitin
- Endomysial fibrosis and fatty replacement
 - reflective of a chronic disease course



Toxic Myopathies

- Toxic myopathies can be caused by
 - prescription or recreational drugs
 - Statins(e.g., atorvastatin, simvastatin, pravastatin)
 - Chloroquine and hydroxychloroquine
 - drug-induced lysosomal storage myopathy
 - ICU myopathy or myosin deficient myopathy
 - corticosteroid therapy
 - certain hormonal imbalances
 - Thyrotoxic myopathy
 - acute or chronic proximal muscle weakness
 - exophthalmic ophthalmoplegia
 - Hypothyroidism
 - cramping or aching of muscles and decreased movement.
 - Reflexes may be slowed.
 - Alcohol
 - acute toxic syndrome of rhabdomyolysis, myoglobinuria, and renal failure
- Morphology
 - fiber atrophy, an increased number of abnormally localized nuclei, glycogen aggregates, and (occasionally) deposition of mucopolysaccharides in connective tissue.



Inherited Diseases of Skeletal Muscle

- Inherited mutations are responsible for a diverse collection of disorders marked by defects in skeletal muscle.
 - In some of these disorders, skeletal muscle is the **main site** of disease
 - in others **multiple organs** are involved (**heart**)
- Inherited myopathies are subdivided into several broad categories based on
 - inheritance pattern
 - anatomic pattern of muscle involvement
 - onset age
 - clinical course
 - underlying pathogenesis

Congenital myopathies

- Typically present in infancy with muscle defects that tend to be **static** or to even **improve over time**.
- They are often associated with distinct structural abnormalities of the muscle.

Muscular dystrophies

- Are characterized by progressive muscle damage
- Typically comes to attention after infancy.
 - Exceptions to this rule are the **congenital muscular dystrophies**
 1. defects in extracellular matrix surrounding myofibers
 2. abnormalities in receptors for extracellular matrix

Muscular Dystrophies

- Include several inherited disorders that have in common
 - Progressive muscle damage
 - Typically manifests itself between childhood and adulthood

X-Linked Muscular Dystrophy with Dystrophin Mutation/ Duchenne and Becker Muscular Dystrophy

- The most common muscular dystrophies
- Stem from mutations that disrupt the function of a large structural protein called **dystrophin(dystrophinopathies)**

A. Duchenne muscular dystrophy

- The most common early onset form
- has a severe progressive phenotype

B. Becker muscular dystrophy

- characterized by later disease onset
- milder phenotype

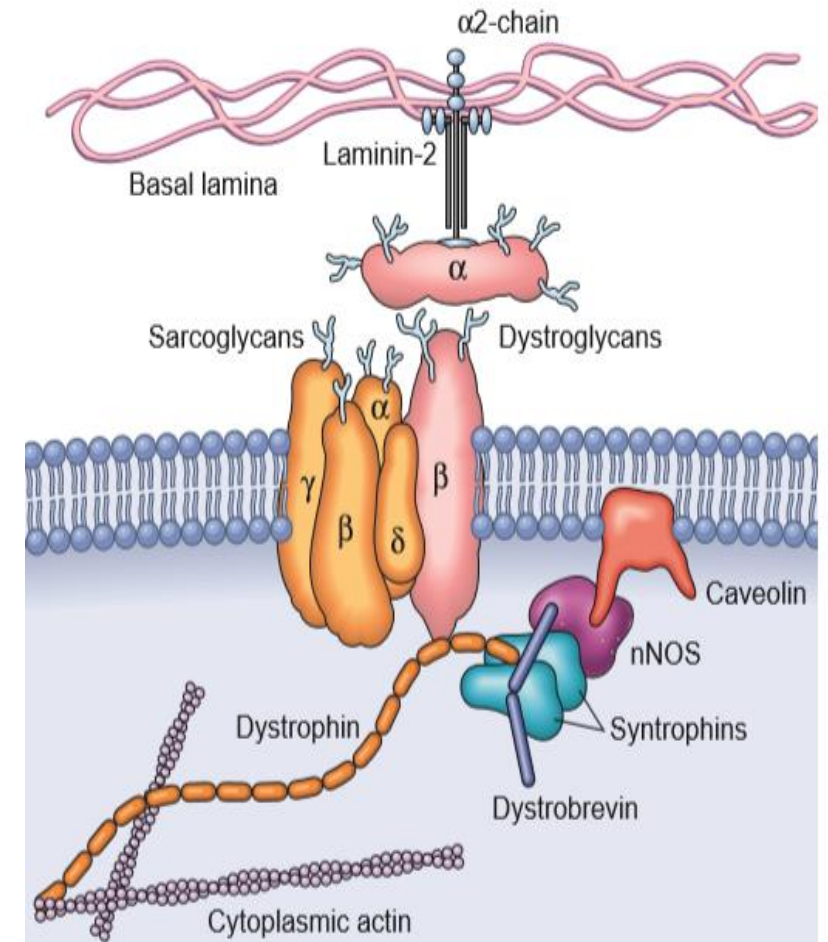
Other rare dystrophinopathies

- present with isolated cardiomyopathy, asymptomatic elevations of creatine kinase levels, or exercise intolerance

❖ Female carriers of dystrophin mutations may be mildly symptomatic due to **unfavorable X-chromosome inactivation**.

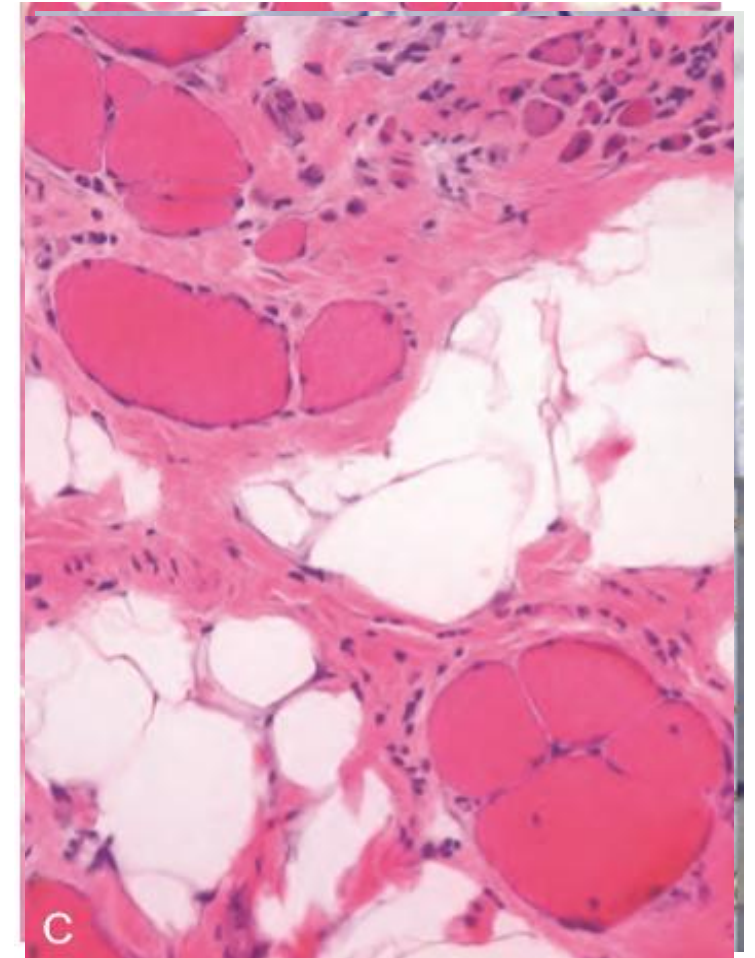
Pathogenesis

- **Duchenne and Becker muscular dystrophy** are caused by
 - Loss-of-function mutations in the **dystrophin gene** on the X chromosome.
 - deletions or frame shift mutations that result in **total absence** of dystrophin=> **DMD**
 - synthesis of a truncated version of dystrophin=> **BMD**
 - Dystrophin provides **mechanical stability** to the myofiber and its cell membrane during muscle contraction
 - Defects in the complex may lead to **small membrane tears** that permit **influx of calcium**, triggering events that result in **myofiber degeneration**



MORPHOLOGY

- Segmental myofiber degeneration and regeneration associated with an admixture of atrophic myofibers
- As the disease progresses
 - muscle tissue is replaced by collagen and fat cells (“fatty replacement” or “fatty infiltration”)
 - Distortion of the fascicular architecture of the muscle
 - IHC
 - absence of the normal sarcolemmal staining pattern in DMD
 - reduced staining in BMD



Clinical Features

- **Duchenne muscular dystrophy**

- normal at birth.
- Very early motor milestones are met, but walking is often delayed
- The first indications of muscle weakness are
 - clumsiness and inability to keep up with peers.
- Weakness begins in the pelvic girdle muscles and then extends to the shoulder girdle
- **Pseudohypertrophy**
 - Enlargement of the muscles of the lower leg associated with weakness
- The mean age of wheel chair dependence is around 9.5 years
- joint contractures, scoliosis, worsening respiratory reserve, and sleep hypoventilation.
- Cardiomyopathy and arrhythmias
- Cognitive impairment
- The mean age of death is 25 to 30 years of age
 - respiratory insufficiency
 - pulmonary infection, or heart failure

- **Becker muscular dystrophy**

- presents in later childhood, adolescence or adult life.
- Its course is more slowly progressive often with a near normal life expectancy



- **Diagnosis**

- Is based on the history, physical exam, and laboratory studies.
- Serum creatine kinase
 - markedly elevated during the first decade of life and then falls as the disease progresses and muscle mass is lost.
- **genetic studies**
 - The presence of a **dystrophin** mutation

- **Treatment**

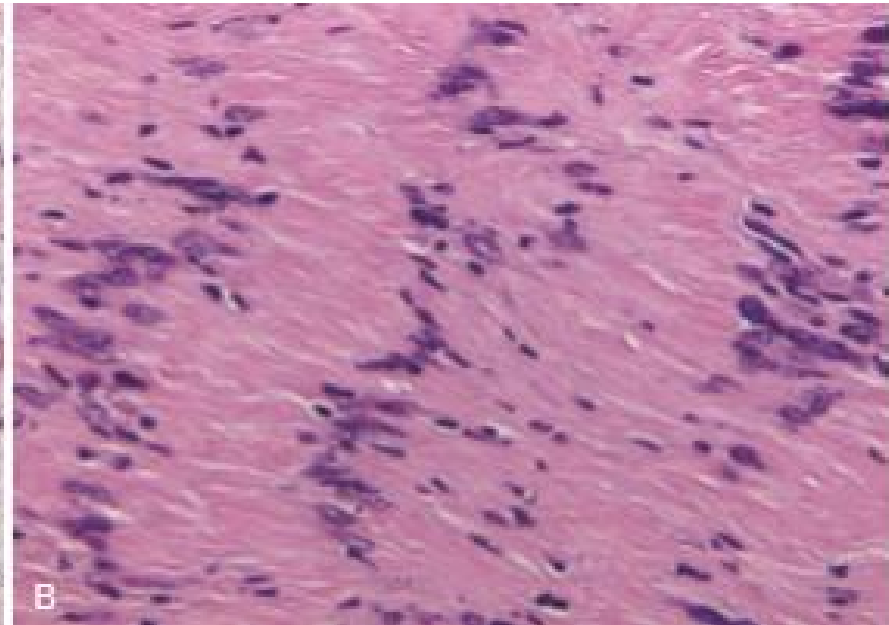
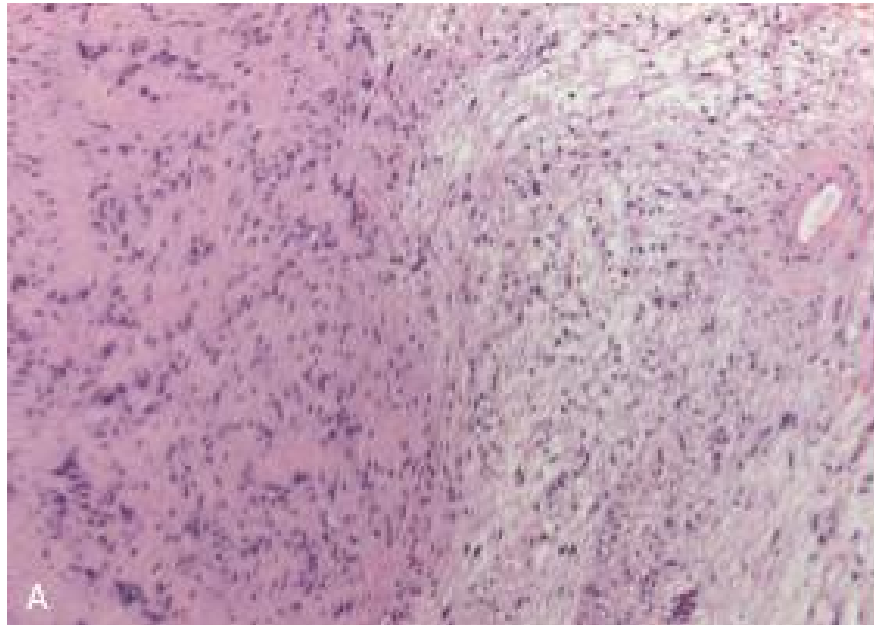
- challenging
- Current treatment consists primarily of supportive care.



PERIPHERAL NERVE SHEATH TUMORS

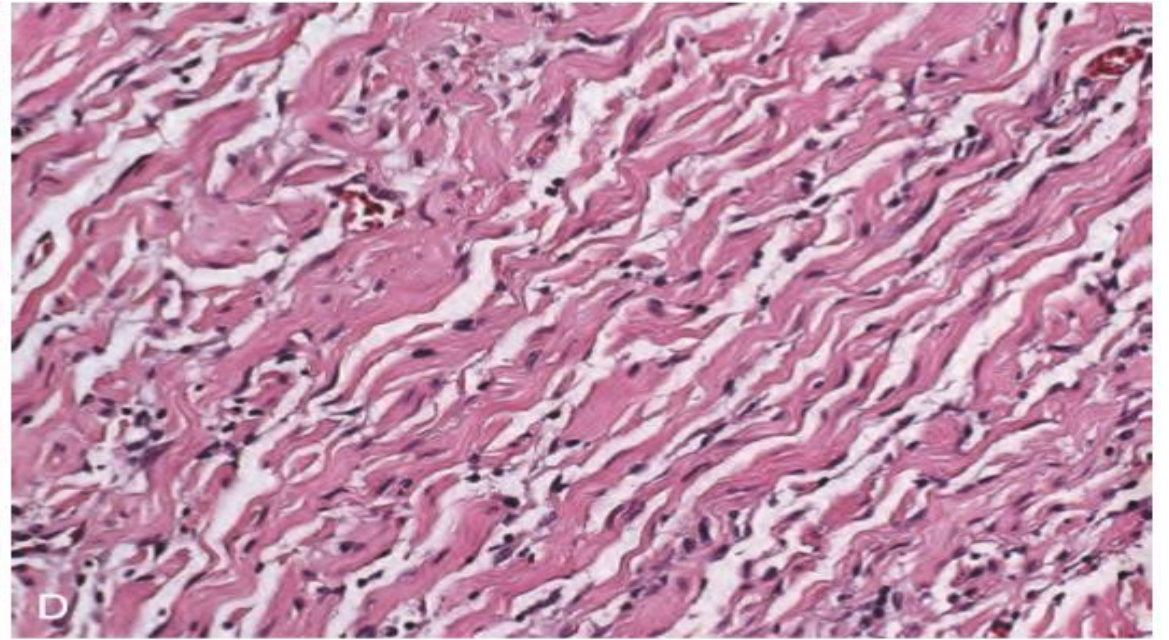
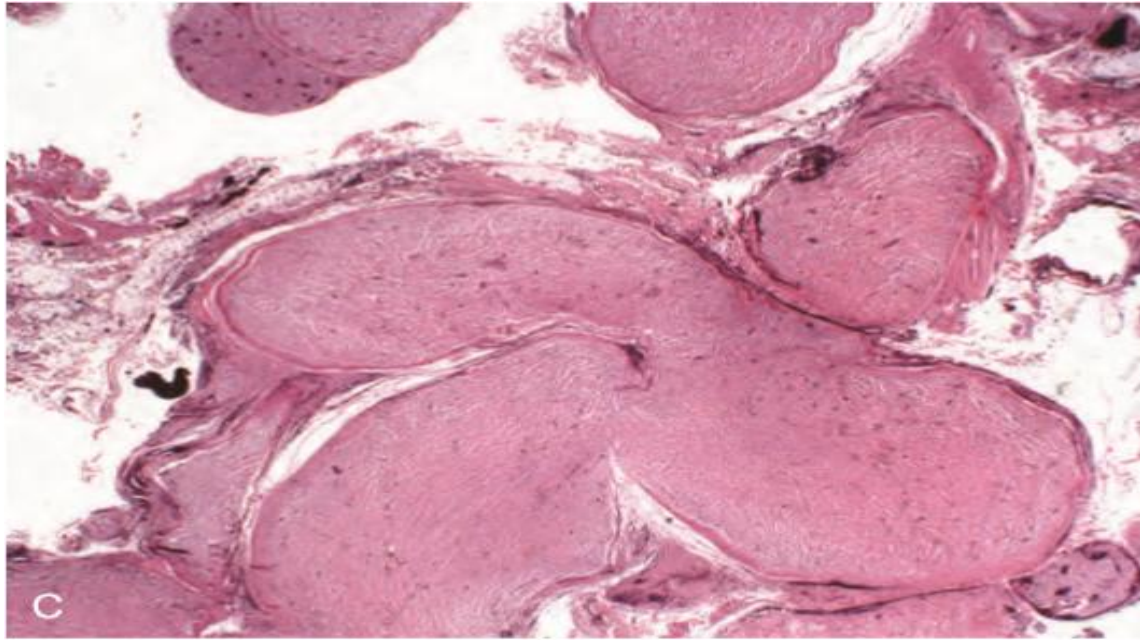
Schwannomas

- benign tumors that exhibit Schwann cell differentiation
- often arise directly from peripheral nerves
- a component of NF2
 - merlin
- well-circumscribed, encapsulated masses
- Microscopically
 - comprised of an admixture of dense and loose areas (Antoni A and Antoni B areas)



Neurofibromas

- benign nerve sheath tumors
- heterogeneous in composition
 - Schwann cells are admixed with perineuriallike cells, fibroblasts, mast cells, and CD34+ spindle cells
- may be either sporadic or NF1-associated
- Different types
 - Superficial cutaneous neurofibromas
 - Diffuse neurofibromas
 - Plexiform neurofibromas
- Complete loss of the NF1 gene product, neurofibromin
- Malignant transformation(5-10%)



Malignant Peripheral Nerve Sheath Tumors

- About half arise in NF1 patients
- Has high grade and low grade variants
- Most are associated with larger peripheral nerves in the chest, abdomen, pelvis, neck, or limb-girdle.
- The lesions are poorly defined tumor masses
- wide range of histologic appearance
- Mitoses, necrosis, and nuclear anaplasia are common
- divergent differentiation

Neurofibromatosis Type 1 and Type 2

Neurofibromatosis Type 1

- A common autosomal dominant disorder
- It is a systemic disease associated with
 - Nonneoplastic manifestations and
 - with a variety of tumors
 - Neurofibromas of all types
 - MPNSTs, gliomas of the optic nerve
 - other glial tumors and hamartomatous lesions, and pheochromocytomas
- Caused by loss-of-function mutations in the NF1 gene

Neurofibromatosis Type 2

- An autosomal dominant disorder
- Results in a range of tumors, most commonly
 - bilateral eighth nerve schwannomas
 - multiple meningiomas, and
 - ependymomas of the spinal cord
- The NF2 gene product, merlin